

IMPACT ANALYSIS OF MULTI-CELL CRASH BOX STRUCTURES WITH TRANSVERSE FUNCTIONALLY GRADED THICKNESS

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APPROVAL PAGE

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ABSTRAK

Selama beberapa dekade terakhir, tren penjualan kendaraan terus meningkat sehingga berbagai perusahaan mobil terus bersaing secara ketat untuk mendesain kendaraan yang terbaik. Meminimalisir berat kendaraan adalah salah satu cara paling efektif untuk meningkatkan daya tarik kendaraan tersebut karena mengurangi bobot kendaraan dapat meningkatkan performa dan mengurangi konsumsi bahan bakar. Namun, mengurangi berat kendaraan harus dilakukan sedemikian rupa agar kendaraan tetap layak tabrak. Dengan meningkatnya tuntutan konstruksi kendaraan ringan, kombinasi konsep *multi-cell* dan *functionally graded thickness* diaplikasikan pada satu struktur *crash box* bernama *functionally graded thickness multi-cell crash box* dimana ketebalan akan divariasikan secara melintang. Memvariasikan ketebalan secara melintang dapat memaksimalkan jumlah material yang mengalami deformasi plastis pada elemen sudut sehingga akan meningkatkan kemampuan menyerap energi struktur tersebut.

Penelitian ini melakukan simulasi numerik menggunakan metode elemen hingga untuk membuktikan kelebihan dari konfigurasi geometri tersebut dalam menyerap energi dari beban impak aksial dan pengaruh variasi *taper ratio* pada performa kelaiktabrakan. Hasil menunjukkan bahwa ada peningkatan hingga **124%** pada SEA, hingga **189,5%** pada MCF dan hingga **83,7%** pada CFE. Namun, terdapat penurunan sebesar **15,8 %** pada MCF dibandingkan dengan crash box *multi-cell* konvensional yang menunjukkan bahwa crash box TFGT *multi-cell* lebih lemah tetapi menawarkan efisiensi berat yang lebih unggul untuk kelaiktabrakannya. Analisis lebih lanjut menunjukkan bahwa ada peningkatan hingga **4,158 %** dalam performa kelaiktabrakan ketika *taper ratio* dibesarkan. CFE paling terpengaruh dengan sensitivitas **0,26** dan MCF paling tidak terpengaruh dengan sensitivitas **0,254**.

Kata kunci: Impak, Kelaiktabrakan, *Crash Box*, Struktur Dinding Tipis, *Multi-Cell*, *Tapered*, *Functionally Graded Thickness*, Parametrik, Sensitivitas

ABSTRACT

Over the last decade, vehicle sales have been constantly increasing despite the fact that recent advancements in information technology drastically reduced the need to travel. As a consequence, car companies aim to design and manufacture the best vehicle possible to compete in the fierce automotive market. Weight saving is one of the most cost-efficient ways to increase the appeal of a vehicle due to the fact that it improves performance and fuel efficiency while reducing the material cost. However, weight reduction should not raise safety concerns by diminishing the vehicle's structural crashworthiness. With increasing demands for a light weight vehicle construction, a combination of both multi-cell and functionally graded thickness concepts are introduced into a single structure. This mixed configuration is termed as functionally graded thickness multi-cell crash box and it is to be tapered transversely. Transverse direction of taper maximizes the amount of material that experiences plastic deformation at the corner sections of the crash box which will result in a higher energy absorbing capability.

This research conducts FEM numerical simulation to establish the advantages of this geometrical configuration in absorbing the energy of axial impact and the influence of taper ratio on the crashworthiness performance. This is achieved by comparing conventional geometries with TFGT multi-cell crash box and analyzing 5 different taper ratios of TFGT multi-cell crash boxes. Results show that there is an improvement of up to **124 %** in SEA, up to **189.5 %** in MCF and up to **83.7 %** in CFE. However, there is a decrease of **15.8 %** in MCF compared to conventional multi-cell crash box which indicates that TFGT multi-cell crash box is less rigid but offers superior weight efficiency in terms of crashworthiness performance. Further analysis indicates that there is an increase of up to **4.158 %** in crashworthiness performance when taper ratio is increased with CFE being the most affected with **0.26** sensitivity and MCF being the least affected with **0.254** sensitivity.

Keywords: Impact, Crashworthiness, Crash Box, Thin-Walled Structures, Multi-Cell, Tapered, Functionally Graded Thickness, Parametric, Sensitivity

Dedicated to Allah SWT, my parents and siblings, my lecturers and advisors, my family and friends, and everyone who benefitted from this undergraduate thesis.

FOREWORD

All praise and gratitude to Allah SWT because of His blessing the author was able to finish this undergraduate thesis with satisfactory results. This undergraduate thesis titled "Impact Analysis of Multi-Cell Crash Box Structures with Transverse Functionally Graded Thickness" is submitted to the Undergraduate Program in Aerospace Engineering of Bandung Institute of Technology as a partial fulfillment of the requirements for the degree of Bachelor of Engineering. In this undergraduate thesis, the author presents a detailed explanation of the research along with the results, analysis, and conclusion.

The author would not have been able to complete this undergraduate thesis without the people who played a role in helping and supporting the author. These people include:

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Bandung, January 15th 2023



Author,

Bayu Dimas Febriansyah

TABLE OF CONTENTS

APPROVAL PAGE	ii
ABSTRAK.....	iii
ABSTRACT.....	iv
FOREWORD	vi
TABLE OF CONTENTS	viii
LIST OF FIGURES.....	x
LIST OF TABLES.....	xiii
1. INTRODUCTION	1
1.1 Research Background.....	1
1.2 Problem Formulation.....	4
1.3 Research Objectives	4
1.4 Scope of Study.....	5
1.5 Research Methodology	5
1.6 Writing System.....	6
2. BASIC THEORY AND LITERATURE	7
2.1 Structural Impact and Crashworthiness.....	7
2.2 Energy Absorbing Strategy.....	10
2.3 Impact Load types	13
2.4 Thin-Walled Structures.....	17
2.5 Transverse Functionally Graded Thickness Multi-Cell Crash Box.....	24
2.6 Trigger	26
2.7 Finite Element Method.....	27
2.8 Parametric Study and Sensitivity Analysis.....	29

3. NUMERICAL MODEL.....	31
3.1 Validation Model	31
3.2 Impact Simulation Design, Parameters, and Condition	34
3.3 Specimen and Impactor Geometries.....	36
3.4 Trigger Application.....	45
3.5 Material	48
3.6 LS-DYNA Keyword Modelling	51
4. RESULT AND ANALYSIS	55
4.1 Validation Model	55
4.2 Conventional Crash Box Geometries	59
4.3 Transverse Functionally Graded Thickness Multi-Cell Crash Box.....	67
4.4 Parametric Study on Taper Ratio Variation	72
4.5 Sensitivity Analysis of Taper Ratio	77
5. CONCLUSION AND FUTURE WORKS	81
5.1 Conclusion.....	81
5.2 Future Works	82
REFERENCES.....	84

LIST OF FIGURES

Figure 1.1 Global car sales statistics 2010-2017. [1]	1
Figure 1.2 Crumple zones in a vehicle. [2]	2
Figure 1.3 (a) multi-corner polygon, (b) multi-cell tubes, (c) transverse functionally graded thickness. [3]	3
Figure 1.4 Transverse functionally graded thickness multi-cell crash box.....	3
Figure 2.1 Vehicle collision.....	7
Figure 2.2 Axial deformation, progressive buckling. [5]	11
Figure 2.3 Crush zones within the frontal structure. [7]	11
Figure 2.4 Crash box location.	12
Figure 2.5 Strain rate sensitivity in 5xxx series aluminum alloys at 338 K. [8]...	13
Figure 2.6 Axial loading. [3]	14
Figure 2.7 Oblique loading. [3].....	15
Figure 2.8 Lateral loading. [3]	16
Figure 2.9 Bending loading. [3]	16
Figure 2.10 Multi-cell structure. [17].....	21
Figure 2.11 Multi-cell simplification. [17].....	22
Figure 2.12 Tapered thin-walled structures. [18]	23
Figure 2.13 Longitudinal functionally graded thickness structure. [19].....	23
Figure 2.14 Functionally graded thickness, (a) in the longitudinal direction and (b) in the transverse direction. [3].....	24
Figure 2.15 Transverse functionally graded thickness multi-cell crash box.....	25
Figure 2.16 Taper ratio. [19]	25
Figure 2.17 Graded thickness distribution. [20].....	26
Figure 2.18 Various types of crash box triggers. [32]	27
Figure 2.19 Finite element model of an aircraft.....	28
Figure 2.20 Illustration of (a) element formulation 1, (b) element formulation 2.	29
Figure 3.1 Instron 5882.....	31
Figure 3.2 Double sided taper TFGT tube, TR=2 (0.8, 1.6).	32
Figure 3.3 ASTM A-36 mild steel stress-strain curve. [20]	33
Figure 3.4 Isometric and front view of the validation model.....	33

Figure 3.5 Top and isometric view of the square tube.	35
Figure 3.6 Initial position of the simulation.	36
Figure 3.7 Conventional square tube crash box.	37
Figure 3.8 4-cell multi-cell crash box.	38
Figure 3.9 Transverse functionally graded thickness crash box.	38
Figure 3.10 TFGT multi-cell crash box with a taper ratio of 2.	39
Figure 3.11 TFGT multi-cell cross-sectional geometries.	40
Figure 3.12 Rigid body impactor top view.	41
Figure 3.13 Rigid body impactor iso view.	41
Figure 3.14 Square crash box mesh.	42
Figure 3.15 4-cell multi-cell crash box mesh.	42
Figure 3.16 Transverse functionally graded thickness crash box mesh.	43
Figure 3.17 TFGT multi-cell crash box A (TR=1.5) mesh.	43
Figure 3.18 TFGT multi-cell crash box B (TR=1.727) mesh.	43
Figure 3.19 TFGT multi-cell crash box C (TR=2) mesh.	44
Figure 3.20 TFGT multi-cell crash box D (TR=2.333) mesh.	44
Figure 3.21 TFGT multi-cell crash box E (TR=2.75) mesh.	44
Figure 3.22 Square tube with trigger.	46
Figure 3.23 Multi-cell tube with trigger.	46
Figure 3.24 TFGT tube with trigger.	46
Figure 3.25 TFGT multi-cell A (TR=1.5) with trigger.	47
Figure 3.26 TFGT multi-cell B (TR=1.727) with trigger.	47
Figure 3.27 TFGT multi-cell C (TR=2) with trigger.	47
Figure 3.28 TFGT multi-cell D (TR=2.333) with trigger.	48
Figure 3.29 TFGT multi-cell E (TR=2.75) with trigger.	48
Figure 3.30 Aluminum 6060-T4 stress-strain curve.	50
Figure 4.1 Validation model deformation.	55
Figure 4.2 Energy plot of the baseline model.	56
Figure 4.3 Force-displacement curve, baseline model.	56
Figure 4.4 Deformation comparison.	57
Figure 4.5 Final deformation state.	58
Figure 4.6 Force-displacement curve, comparison.	58

Figure 4.7 Plain square tube deformation.	60
Figure 4.8 Energy plot of plain square tube.	60
Figure 4.9 Force-displacement curve, plain square tube.	61
Figure 4.10 4-cell multi-cell tube deformation.	62
Figure 4.11 Energy plot of 4-cell multi-cell tube.	62
Figure 4.12 Force-displacement curve, 4-cell multi-cell tube.	63
Figure 4.13 Transverse functionally graded thickness tube deformation.	64
Figure 4.14 Energy plot of TFGT square tube.	65
Figure 4.15 Force-displacement curve, TFGT square tube.	65
Figure 4.16 Force-displacement curve, conventional tubes.	66
Figure 4.17 Transverse functionally graded thickness multi-cell crash box deformation.	68
Figure 4.18 Energy plot of TFGT multi-cell crash box.	68
Figure 4.19 Force-displacement curve, TFGT multi-cell crash box.	69
Figure 4.20 Force-displacement curve, TFGT multi-cell vs conventional.	70
Figure 4.21 TFGT multi-cell crash box deformation, TR=1.5.	72
Figure 4.22 TFGT multi-cell crash box deformation, TR=1.727.	72
Figure 4.23 TFGT multi-cell crash box deformation, TR=2.	73
Figure 4.24 TFGT multi-cell crash box deformation, TR=2.333.	73
Figure 4.25 TFGT multi-cell crash box deformation, TR=2.75.	74
Figure 4.26 Force-displacement curve, taper ratio variation.	74
Figure 4.27 SEA vs taper ratio curve.	77
Figure 4.28 MCF vs taper ratio curve.	78
Figure 4.29 CFE vs taper ratio curve.	78

LIST OF TABLES

Table 3.1 AA 6061 O mechanical properties. [20].....	32
Table 3.2 Combined simulation parameters.....	35
Table 3.3 Geometrical parameters for TFGT multi-cell crash boxes.....	40
Table 3.4 Half folding lengths of TFGT multi-cell crash boxes.	45
Table 3.5 Aluminum 6060-T4 composition.	49
Table 3.6 Aluminum 6060-T4 mechanical properties.....	49
Table 3.7 Aluminum 6060-T4 stress-strain data.	50
Table 3.8 Impactor properties.	51
Table 4.1 Crashworthiness indicators of baseline model.	57
Table 4.2 Results comparison.	59
Table 4.3 Crashworthiness indicators of plain square tube.....	61
Table 4.4 Crashworthiness indicators of 4-cell multi-cell tube.....	63
Table 4.5 Crashworthiness indicators of TFGT square tube.	66
Table 4.6 Crashworthiness indicators, conventional tubes.	67
Table 4.7 Crashworthiness indicators of TFGT multi-cell crash box.	69
Table 4.8 SEA improvement.	70
Table 4.9 MCF improvement.	71
Table 4.10 CFE improvement.	71
Table 4.11 Crashworthiness indicators of varying taper ratios.	75
Table 4.12 SEA improvement.	75
Table 4.13 MCF improvement.	76
Table 4.14 CFE improvement.	76
Table 4.15 Sensitivity of SEA to taper ratio.....	79
Table 4.16 Sensitivity of MCF to taper ratio.....	79
Table 4.17 Sensitivity of CFE to taper ratio.....	80

CHAPTER I

INTRODUCTION

1.1 Research Background

Motorized road vehicles such as cars, trucks, and buses, are the most commonly used mode of transportation in the world. Cars especially has been in widespread use after its invention in the late 19th century due to the fact that it can be purchased and driven by everyone. It has been the preferred method of transport for short to medium distances, and sometimes for long distances if the destination is accessible by road. In developing countries, cars are particularly sought after as public transportation is still flawed.



Figure 1.1 Global car sales statistics 2010-2017. [1]

Over the last decade, car sales have been constantly increasing despite the fact that recent advancements in information technology drastically reduced the need to travel as illustrated by Figure 1.1. As a consequence, car companies aim to design and manufacture the best vehicle possible (within its category) to compete in the fierce automotive market. Weight saving is one of the easiest ways to increase the appeal of a vehicle. Reducing a vehicles weight can greatly improve performance and fuel consumption. However, weight reduction should not raise safety concerns by reducing the vehicle's structural crashworthiness.

The safety of a vehicle is determined mostly by its structural crashworthiness. In the event of a high velocity frontal impact, the vehicle's crumple zones are expected to absorb the massive amount of kinetical energy while still maintaining a reasonable deceleration as to not injure the occupants, which is shown in Fig. 1.2. This is achieved by incorporating a crash box structure in the primary crush zone of the crumple zone that is designed to transform the kinetical energy to potential energy by allowing its material to deform progressively. The efficiency of this energy transformation is determined by its deformation mode, which is affected by the geometrical design of the crash box, the material, and also the direction of impact.

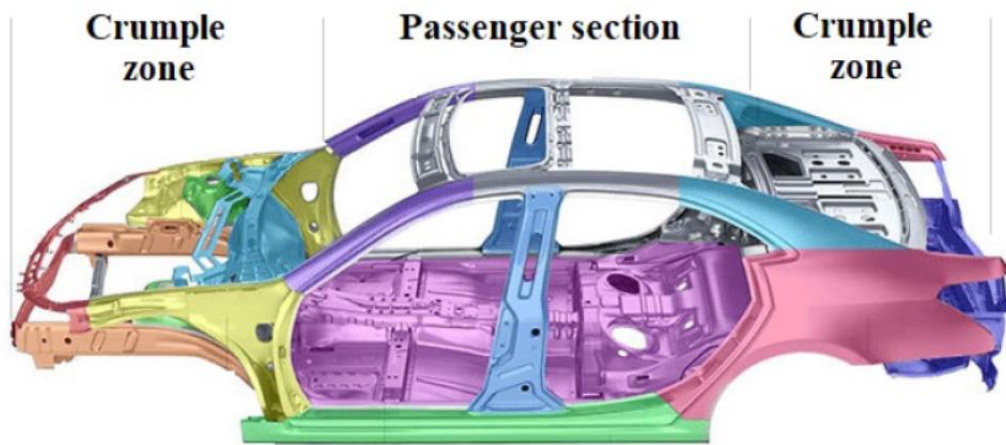


Figure 1.2 Crumple zones in a vehicle. [2]

Thin-walled structures is the most popular form of collapsible energy absorbers [3]. This is most likely due to its performance when subjected to dynamic impact loading combined with its practicality. It is cheap and easy to manufacture while also being light weight which is beneficial to the vehicle's performance. There are wide variations of geometric shapes that serves as a basis for the thin-walled structure that could be applied to a crash box's cross-section namely triangle, square, polygons, and a circle, Fig. 1.3. These shapes may also be designed with a multi-cell configuration to increase the amount of corner element or even tapered to add more material in the desired location which is often called a functionally graded thickness.

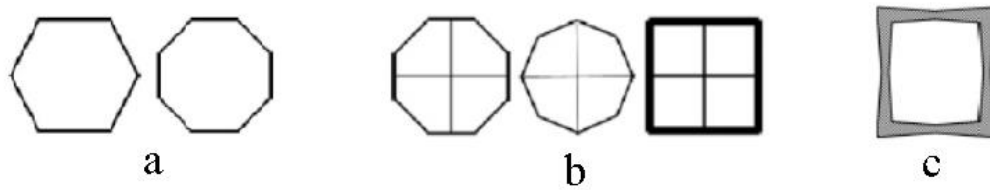


Figure 1.3 (a) multi-corner polygon, (b) multi-cell tubes, (c) transverse functionally graded thickness. [3]

To satisfy with the demands of a light weight crash box, a combination of both multi-cell and functionally graded thickness concepts are combined into a single structure. This relatively new geometrical configuration is called transverse functionally graded thickness multi-cell crash box because the wall thickness is tapered transversely as shown in Fig. 1.4.

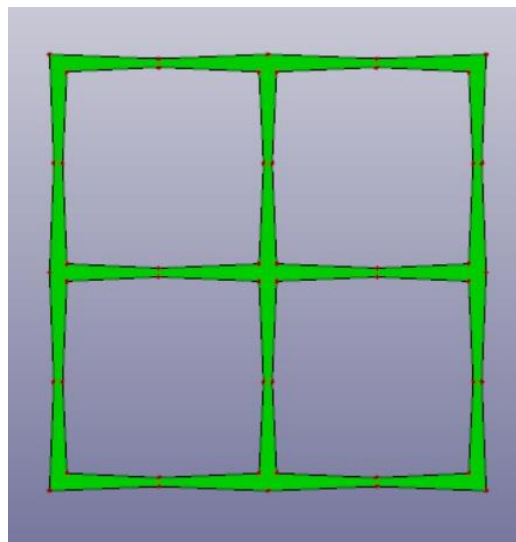


Figure 1.4 Transverse functionally graded thickness multi-cell crash box.

Besides increasing the number of corners by incorporating cells, this crash box also adds material around the corner elements in order to maximize the amount of material that experiences plastic deformation which will result in a better energy absorbing capability. This research aims to establish the advantages of this geometrical configuration in absorbing the energy of axial impact for application as a crash box in the frontal structure of a vehicle. Additionally, this research also conducts parametric study and sensitivity analysis on taper ratio variation to establish the relationship between taper ratio and crashworthiness performance.

1.2 Problem Formulation

Following the previously mentioned research objectives, the following are the problem formulations for this research:

1. What happens when transverse functionally graded thickness multi-cell crash box is subjected to axial impact load.
2. What is the crashworthiness performance of transverse functionally graded thickness multi-cell crash box?
3. How applying transverse functionally graded thickness on the multi-cell crash box affect the crashworthiness performance?
4. How varying the taper ratio of transverse functionally graded thickness multi-cell crash box affect the crashworthiness performance?
5. How sensitive is taper ratio variation on the crashworthiness performance of transverse functionally graded thickness multi-cell crash box?

1.3 Research Objectives

Based on the problem formulation, the purpose of this research is to achieve the following objectives:

1. To analyze the deformation behavior of transverse functionally graded thickness multi-cell crash box when subjected to axial impact load.
2. To obtain the crashworthiness performance of transverse functionally graded thickness multi-cell crash box.
3. To compare the specific energy absorption of the transverse functionally graded thickness multi-cell crash box with conventional geometries.
4. To determine the influence of taper ratio on the transverse functionally graded thickness multi-cell crash box.
5. To determine the sensitivity of taper ratio variation on the crashworthiness performance of transverse functionally graded multi-cell crash box.

1.4 Scope of Study

To limit the extent of this research, the fulfillment criteria is based on the scope that is determined by these boundaries:

1. The material used on the transverse functionally graded thickness multi-cell crash box and other crash box specimens is aluminum Al6060-T4.
2. The analysis will be conducted by a numerical simulation using finite element method (FEM).
3. The numerical simulation will use LS-DYNA as the FEM software.
4. The analysis will be conducted on the transverse functionally graded thickness multi-cell crash box by comparing it with other crash box configurations that is simulated with the same simulation parameters.

1.5 Research Methodology

Research for this undergraduate thesis is conducted with these methodology guidelines:

1. Literature study, which is conducted by reading and reviewing published papers, books, and past final assignments that is related to this research.
2. Analytical modelling, which is conducted by simplifying the research scope and boundaries in order to approximate the simulation.
3. Geometrical design, which is conducted using a computer-aided design (CAD) software to accurately model the desired geometry.
4. Numerical simulation, which is conducted using the finite element method (FEM) software LS-DYNA with specific simulation parameters which is based on the research scope and boundaries.
5. Post-processing, which is conducted using the same finite element method (FEM) software LS-Prepost and Microsoft Excel to analyze the result obtained from the numerical simulation.
6. Undergraduate thesis writing, which is conducted using Microsoft Word using previously processed data from Microsoft Excel.

1.6 Writing System

This undergraduate thesis will be divided into 5 chapters incorporating the writing system as follows:

CHAPTER I INTRODUCTION

This chapter contains the background, problem formulation, research objectives, research boundaries, research methodology, and writing system of this final assignment proposal.

CHAPTER II BASIC THEORY AND LITERATURE

This chapter contains the theory behind this research based on published papers, books, and past final assignments.

CHAPTER III NUMERICAL MODEL

This chapter contains detailed information of the numerical modelling process which is configured based on simulation parameters obtained from various literature and studies that is suitable for this research.

CHAPTER IV RESULT AND ANALYSIS

This chapter contains the processed results of the numerical simulation along with the analysis of those results in order to answer the research problem and fulfill the research objectives. This chapter also contains the sensitivity analysis of taper ratio variation on TFGT multi-cell crash with trigger.

CHAPTER V CONCLUSION AND FUTURE WORKS

This chapter contains the conclusion of the analyzed results of the numerical simulation along with suggestions for improvement and future works.

CHAPTER II

BASIC THEORY AND LITERATURE

2.1 Structural Impact and Crashworthiness

Impact is generally defined as a large dynamic, or even explosive loads that occurred between two or more bodies which results in permanent (inelastic) deformation as a mean to absorb the impact energy [4]. When a structure experiences this type of loading, its behavior during deformation depends on the design of the structure itself, which indicates that a well-engineered structure could achieve desirable deformation when subjected to impact. Structural impact studies this behavior of structures during impact and how it should be implemented for safety and risk management on structures used by humans.



Figure 2.1 Vehicle collision. [35]

Large dynamic impacts on a land vehicle usually occurs during a high velocity collision. Countless measures have been taken to prevent road accidents; but as with other mode of transportation, it may still occur as a result of human error, mechanical failure, poor environmental conditions, etc. If the vehicle is travelling above a certain velocity, these accidents could be dangerous for the occupants since the outcome would most likely be crashing into another object that is within its path which induces an impact load onto the structure of the vehicle, Fig. 2.1.

Crashworthiness is a term used to define a measure of the ability of a structure and any of its components to protect the occupants in survivable crashes. In a vehicle, this is achieved by designing the structure so that it deforms plastically in the desired manner in order to absorb as much impact energy as possible while still preserving sufficient space around the occupants to limit bodily injuries [5]. Evaluation for crashworthiness is conducted using both analytical and experimental methods with the ultimate goal of occupant's safety, low risk of injury or death, and the protection of occupants in the event of an accident.

When discussed further there are a few crashworthiness requirements that needs to be taken into consideration when designing a vehicle. The structure should be relatively stiff in bending and torsion to maintain a comfortable ride and precise handling but still provide sufficient energy absorbing capabilities during an accident. In detail, the requirements for vehicle structures are as follows:

- The frontal structure of the vehicle should be deformable but still relatively stiff. It should include a dedicated crumple zone to absorb energy by plastically deforming and also to prevent the intrusion of vehicle components or foreign objects into the occupant compartment which may cause physical injuries. The front of the vehicle should also be adequately long to house the frontal structure along with its crumple zone which indicates that vehicles with short frontal ends are harder to be designed crashworthy.
- The rear structure of the vehicle should be deformable in order to maintain the integrity of the rear occupant compartment and also the fuel tank (if located near the rear).
- The side structures and doors should be able to minimize intrusion to the occupant compartment during a side impact. The doors should also be prevented from opening under impact loads.
- The roof structure should be stiff and strong enough to maintain its integrity during a rollover.

As evident from the requirements above, the most complex requirement is the one related to the frontal structure. This is due to the fact that during regular usage vehicles will always travel forward, with reverse only being used to maneuver, meaning that during an accident the vehicle will most likely experience impact on the frontal part because it is moving with velocity on that direction.

Zang et al [6] uses a few crashworthiness indicators in order to properly evaluate the crashworthiness of the energy-absorbing structures, which in this case is the frontal structure. These indicators are the Energy Absorption (EA), the Specific Energy Absorption (SEA), the Initial Peak Force (Fmax), the Mean Crushing Force (Fmean / MCF), and the Crushing Force Efficiency (CFE).

The EA is the total energy absorbed by the structure during the deformation process which could be calculated by integrating the load-displacement curves. The EA equation is defined as follows,

$$EA = \int_0^d F(x)dx \quad (1)$$

where $F(x)$ is the crushing force, and d is the crushing displacement.

The SEA is the ratio of the total energy absorbed by the structure to the mass of the structure. It is one of the most important indicators when evaluating the crashworthiness of a structure because weight is a great consideration when designing a structure, especially in vehicles. The SEA equation is as follows,

$$SEA = \frac{EA}{M_{deformed}} \quad (2)$$

where EA is the Energy Absorption while $M_{deformed}$ is the mass of the deformed structure. In most cases, it is desirable to have the value of SEA as large as possible as it indicates the better crashworthiness performance for its weight.

The Fmax is the highest crushing force during the deformation process which usually happened early in the process at the critical state of the structure. Its value is able to represent the injury severity that is brought to the vehicle's occupants during an accident. A large value of Fmax indicates a high deceleration,

which would result in more severe injuries. This suggest that a large value of F_{max} should be avoided when designing the energy-absorbing structure of a vehicle with occupants.

The F_{mean} is the ratio of the total energy absorbed by the structure to the displacement. Its value shows the overall energy absorption capability. The F_{mean} equation is defined as follows,

$$F_{mean} = MCF = \frac{EA}{d} \quad (3)$$

where EA is the Energy Absorption while d is the crushing displacement. If the crushing displacement of different structures is identical or incredibly similar, the structure with the higher F_{mean} will be able to absorb more energy.

The CFE is the ratio of the mean crushing force to the highest crushing force during the deformation process. The CFE equation is defined below.

$$CFE = \frac{F_{mean}}{F_{max}} \quad (4)$$

In general, the value of F_{mean} will always be lower than F_{max} which means that the value of CFE is most likely going to be under 1. The closer the value is to 1 suggests that the highest crushing force is approaching the mean crushing force, indicating a more stable load. As a result, in order to avoid severe injuries to the vehicle's occupants, the value of CFE should be as large as possible.

2.2 Energy Absorbing Strategy

Traditional frontal impact energy-absorbing structure deforms in a mixed axial and bending mode, with the bending mode being more dominant on the deformation because of its lower energy content [5]. This indicates that bending is the far less efficient mode for energy absorption and will cause designs to be heavier whereas axial mode will be more suitable because of its lighter weight. If the main energy-absorbing structure is designed properly for axial mode deformation, it will deform with a relatively uniform, progressive structural collapse. This type of deformation is called progressive buckling as illustrated in Fig. 2.2.



Figure 2.2 Axial deformation, progressive buckling. [5]

As mentioned before, only the main energy-absorbing structure should deform in this manner and other components within the frontal structure should be designed according to its function. Three distinct crush zones are defined for the frontal structure and is comprised of a soft zone, a primary crush zone, and a secondary crush zone, Fig. 2.3. A soft zone is generally designed to protect humans (e.g., pedestrians, cyclists, etc.) if they crashed into the vehicle and also for low velocity, vehicle-to-vehicle collisions. This is usually installed on the vehicle in the form of a bumper. A primary crush zone is the main energy-absorbing structure of the vehicle that is designed to buckle progressively when subjected to axial impact loads during medium and high velocity accidents. The structure that made up the primary crush zone is often called a crash box and it spans from the end of the bumper until the front of the engine as shown in Fig. 2.4. The last zone is the secondary crush zone which is the structure that will serve as a barrier between the main energy absorber and the occupant compartment. This zone should prevent the intrusion of deformed vehicle components and foreign objects into the occupant compartment.

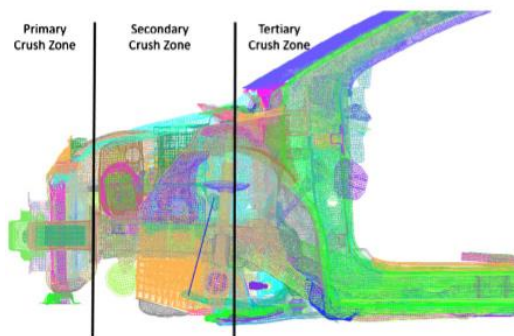


Figure 2.3 Crush zones within the frontal structure. [7]

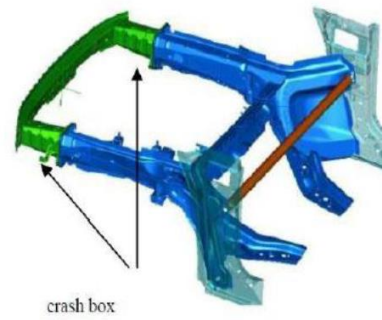


Figure 2.4 Crash box location. [7]

Crash box's design and material of are crucial in determining how it deforms, which will affect the vehicle's crashworthiness performance. A well-designed crash box geometry will allow massive amounts of energy to be absorbed while maintaining its weight to a minimum. Crash boxes usually incorporates a thin-walled structure which has a vast number of geometrical configurations and design choices that will be further discussed in Section 2.4. Material choice is also as important because besides having unique density and strength, different materials also have different ductility which affects the way the crash box deforms. Crash boxes are mostly manufactured using metallic materials such as aluminum and steel, with aluminum offering a significant advantage when pursuing a lightweight construction.

According to Baroutaji et al [3], metallic materials are sensitive to loading rate which indicates that the mechanical properties of the crash box under dynamic loading are different when compared to the same crash box under quasi-static loading. This sensitivity to loading rate is an independent material property separate from the geometrical property of the crash box, meaning that a material's sensitivity to loading rate cannot be altered by changing the geometry, Fig. 2.5.

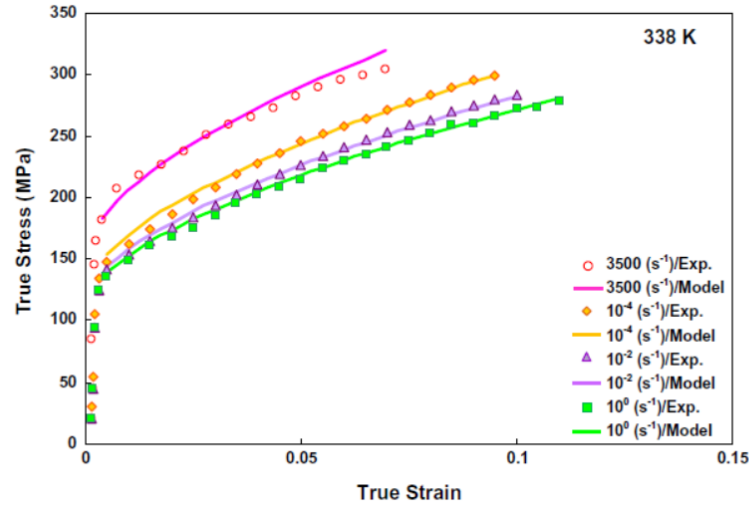


Figure 2.5 Strain rate sensitivity in 5xxx series aluminum alloys at 338 K. [8]

The load that the crash box will experience during a frontal impact is considered a dynamic loading, which means that it is beneficial for metallic materials to have a strain rate sensitivity behavior because it allows the material to achieve a greater energy absorption capacity relative to its performance on quasi-static loading. In practice, strain rate effects are usually considered by using a dynamic plastic flow stress instead of static plastic flow stress. The dynamic plastic flow stress could be calculated using the Cowper-Symonds equation as follows,

$$\sigma_d = \sigma_s \left(1 + \left(\frac{\dot{\epsilon}}{D} \right)^{\frac{1}{q}} \right) \quad (5)$$

where σ_s is static yield stress, $\dot{\epsilon}$ is the strain rate, and D and q being the parameters of the material.

2.3 Impact Load types

Although the motion of an impact is straightforward, depending on the orientation of the structure that is being subjected to impact, different crashworthiness performance may occur, which indicates that the specific way it is subjected to impact loading are a crucial parameter. This is due to how the cross-section of the structure normal to the load direction vary depending on the

orientation, which determines the amount of the structure's material that will experience plastic deformation and contribute in the energy absorption process. The scope of this research only covers perfectly axial loading. However, an understanding of other load types and how they affect the energy absorption and crashworthiness of different structures will help the design and optimization of the crash box.

2.3.1 Axial Loading

Axial loading occurred when the impact load direction is normal to the cross-section of the structure, Fig. 2.6.

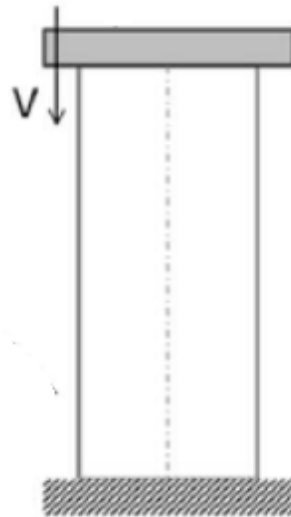


Figure 2.6 Axial loading. [3]

This is the main type of impact load a crash box should withstand as the crash box is generally designed to absorb energy during a frontal impact. Alghamdi [9] states that the specific energy absorbing capacity of axially loaded structures is approximately 10 times greater than that in laterally loaded structures because it allows most of the structure's material to deform plastically and progressively buckle in the intended direction. This is the reason why most crash boxes are designed and oriented for axial loading in the event of a frontal impact.

2.3.2 Oblique Loading

Oblique loading occurred when the impact load direction is at an angle to the cross-section of the structure and is also the other type of load a crash box should withstand, Fig. 2.7. This is considered as the most realistic type of loading as most real accidents will not perfectly align with the crash box for a perfectly axial loading, which is why according to Reyes et al [10] the automotive industry requires that the bumper and crash box system of a vehicle should be able to withstand loads applied in a 30° angle to the longitudinal axis.

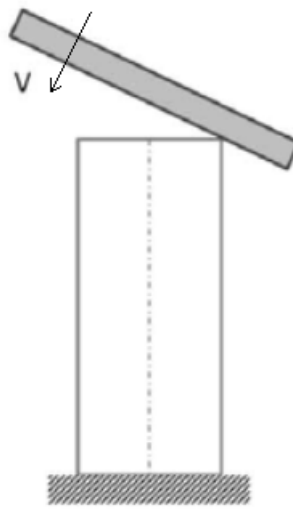


Figure 2.7 Oblique loading. [3]

Generally, this type of loading causes the structure to bend while experiencing some form of progressive folding. The energy absorption capability and the peak force of the structure will decrease while the tendency to undergo global bending will increase when increasing the loading angle.

2.3.3 Lateral Loading

Lateral loading occurred when the impact load direction is perpendicular to the longitudinal axis of the structure and is applied throughout the length of the structure as shown on Fig. 2.8.

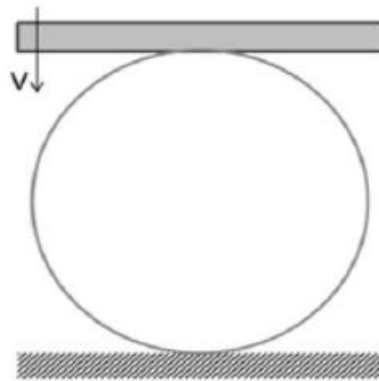


Figure 2.8 Lateral loading. [3]

During a frontal impact, this type of loading is usually withstood by the cross-beam of the bumper. The deformation different from other types of loading and often called the flattening process.

2.3.4 Bending Loading

Bending loading occurred when the impact load direction is perpendicular to the longitudinal axis of the structure, similar to lateral loading. However, instead of applying it throughout the length of the structure the load is only applied on a point somewhere on the longitudinal axis, within the length of the structure, Fig. 2.9.

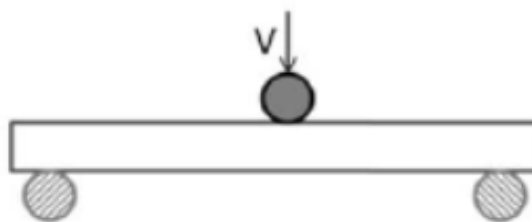


Figure 2.9 Bending loading. [3]

During a frontal impact, this type of loading is mainly withstood by the cross-beam of the bumper but it may also be subjected to the secondary crush zone, the roof pillars, and the side-door beams. The deformation starts with a local indentation at the point where the load is applied and then the structure will experience global bending due to substantial reduction of the structure's cross-sectional area at that point.

2.4 Thin-Walled Structures

Thin-walled structures are generally considered as an engineering construction method in which the internal section of the component is hollow, leaving the remaining thin material around the void as the load bearing structure. It is one of the most efficient methods to shave weight from a structure while retaining most of the structural integrity and is applied on increasingly diverse fields, ranging from vehicles to buildings. Thin-walled structures that is applied with axial loads are often referred to as tubes whereas for bending loads it is called a beam.

The main energy-absorbing component of a vehicle, or a crash box, which is situated at the primary crush zone of a vehicle's frontal structure, often uses thin-walled structures because it is the most popular form of collapsible kinetic energy absorbers [3]. When incorporated as a crash box, the overall crashworthiness of the frontal structure will improve drastically while maintaining a low weight. In addition, thin-walled structures are also cost-effective and easy to manufacture which results in high efficiency.

The crashworthiness performance and energy absorption capabilities of thin-walled structures depends on its material, loading type, and structural geometry. The effects of materials and loading types has been stated on previous sections so this section will only discuss the influence of geometry on axially loaded thin-walled structures (a crash box). There are wide variations of geometric shapes that could be applied to a crash box's cross-section namely triangle, square, polygons, and a circle. These shapes may also be designed with a multi-cell configuration to increase the amount of corner element or even tapered to add more material in the desired location.

2.4.1 Conventional

Conventional thin wall geometry purely extrudes the basic shape of the cross-section in uniform thickness without additional internal geometries which results in a single volume, hollow interior. As stated before, basic shapes that could be incorporated into a thin-walled structures includes triangles, squares, polygons, and circles. More complex shapes could still be considered conventional such as rounded polygons (using corner fillets) or an abstract geometry as long as the hollow interior is a single volume and the wall thickness is uniform. For this research, a uniform thickness multi-corner structure will also be considered conventional as it is basically an extruded irregular polygon.

2.4.1.1 Triangular

Triangular thin-walled structure has the least energy absorption capacity among other polygons due to the fact that it has the fewest corners. Fan et al [11] conducted research on equilateral triangular structures under quasi-static loading and found that it collapses in two main deformation modes depending on its geometrical dimensions. In structures with $c/t > 60$ the deformation is in rotational symmetrical mode whereas in structures with $c/t \leq 55.3$ the deformation is in diamond mode. The theoretical model for rotational symmetrical mode is defined as follows,

$$MCF = 0.25\sigma_o t^2 (12.038 + 17.126\sqrt{clt}) \quad (6)$$

and the theoretical model for diamond mode is defined as follows,

$$MCF = 0.25\sigma_o t^2 (15.561 + 15.052\sqrt{clt}) \quad (7)$$

where t and c are the thickness and side length of the triangular structure. As a result of its low energy absorbing capabilities and poor crashworthiness, this geometrical configuration did not receive much attention in the academic field and it is rarely used in vehicles.

2.4.1.2 Square

Square thin-walled structure has an additional corner when compared to a triangular structure. The cross-section dimension of a square structure is an integral factor that determines its deformation mode and energy absorbing capabilities. Depending on the width to thickness ratio, the deformation mode could be either extensional, inextensional, or mixed [4]. Abramowicz and Jones [12] proposes that the mean crushing force of axially loaded square structures, after accounting the strain hardening effect, is defined as follows,

$$MCF = 13.06\sigma_o b^{1/3} t^{5/3} \quad (8)$$

where b and t are the width and thickness of the square structure. Unlike its triangle counterpart, this geometrical configuration is well researched and commonly used in vehicle structures due to its practicality. Square structures are very easy and cheap to manufacture, offer reasonable structural strength and energy absorbing capabilities, and it is easy to mount to other components or structural members.

2.4.1.3 Polygon

Polygon thin-walled structures, which includes multi-corner geometries, offers high energy absorbing capabilities due to the fact that it has a lot of corners. In triangular and square structures, severe deformation occurs in the area near the corner, meaning that adding corners to a structure will increase the amount of material that experiences plastic deformation which leads to a higher energy absorbing capability. The deformation mode is similar to square structures, namely extensional, inextensional, and mixed. However, in polygon and multi-corner structures, it depends on the angle of the corner, not just the side width and thickness [3]. Acute angles deform with inextensional mode whereas obtuse angles above 120° deforms with extensional mode. Mixed deformation mode occurs for angles between 90° and 120° which is also the range where the highest energy absorption efficiency is achieved.

Theoretical calculation for mean crushing force is also possible for geometries with multiple corners. According to Abramowicz and Wierzbicki [13], the mean crushing forces of hexagonal tubes are defined as follows

$$MCF = 20.23\sigma_o b^{0.4} t^{1.6} \quad (9)$$

where b and t are the side length and thickness of the hexagonal tube. It is evident from the equation that hexagonal tubes have a higher mean crushing force than a square structure with the same cross-sectional area, which confirms that adding corners to a structure will increase its energy absorbing capabilities. However, Yamashita et al [14] observed that improvements are negligible for structures with number of corners above 11. This is possibly due to the fact that corner angle will also increase with the increase of number of corners and as the number of corner increases, the corner angle will deviate away from the optimal range.

2.4.1.4 Circle

Circular thin-walled structures have no corner sections and is uniformly curved throughout its perimeter. When subjected to axial loads, both the ratio of diameter to thickness (D/t) and the ratio of length to thickness (L/t) determines whether the circular structure deforms in axisymmetric mode (ring), non-symmetric mode (diamond), or mixed mode. Diamond mode occurs when $(D/t) > 80$ whereas ring mode occurs when $(D/t) < 50$ and $(L/t) < 2$. For $(D/t) < 50$ and $(L/t) > 2$, mixed mode will occur. Guillo et al [15] developed an equation to calculate the mean crushing force for circular tubes that will work for every deformation mode which is defined as follows

$$MCF = 18.075\sigma_o t^2 \left(\frac{D}{t}\right)^{0.32} \quad (10)$$

where t is the thickness and D is the diameter of the circular tube. Unique to circular tubes, the impact velocity also affects the deformation mode. Wang et al [16] discovered that at relatively low velocities, thin circular tubes will experience a dynamic progressive deformation mode in the form of folds. For all thickness, at medium velocities, the tube will experience end mushrooming with folds forming

at a distance. At high velocities, thick-walled tubes will experience mushrooming and wrinkling. Despite its relatively good energy absorbing capabilities, the use of circular tubes in vehicles are limited due to issues associated with its shape specifically difficulties in mounting to other components or structural members.

2.4.2 Multi-Cell

Section 2.4.1.3 discussed how the number of corner elements on a structure's cross section affect the energy absorbing capability. With this fact, a thin-walled structure with multiple cells within its wall could be designed with the intention of weight-efficient energy absorption as shown in Fig. 2.10. In addition to being weight efficient, a multi-cell construction allows the structure to have multiple corners internally while maintaining a conventional external geometry which increases its practicality.

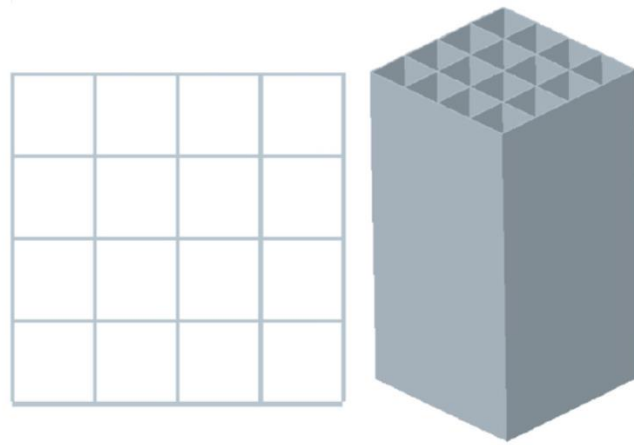


Figure 2.10 Multi-cell structure. [17]

The theoretical calculation for mean crushing force of a multi-cell structure is based on the super folding element method developed by Wierzbicki and Abramowicz. Zhang et al [17] worked based on this method to derive the mean crushing force and the folding length of a multi-cell structure. The multi-cell cross section is simplified into three basic components namely the corner part, the crisscross part, and the T shaped part which allows them to analyze the energy dissipation over the collapse of a fold, Fig. 2.11.

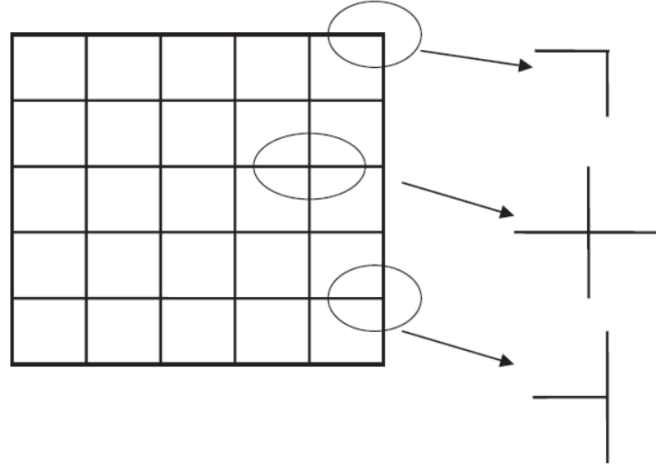


Figure 2.11 Multi-cell simplification. [17]

The folding length and the mean crushing force of multi-cell structures are defined in the equations as follows,

$$H = \sqrt{\frac{\pi L_c t}{2(N_c + 4N_0 + N_T)}} \quad (11)$$

$$MCF = \sigma_o t \sqrt{(N_c + 4N_0 + N_T) \pi L_c t} \quad (12)$$

where H is the folding length, MCF is the mean crushing force, N_c is the number of corners, N_0 is the number of crisscrosses, N_T is the number of T shape, and $L_c t$ is the material area of the cross-section.

2.4.3 Functionally Graded Thickness

Functionally graded thickness is a concept that aims to change the material distribution of a structure in order to increase the energy absorption capabilities without increasing the weight. This is achieved by increasing the amount of material around the area where severe plastic deformation occurs so that more material will experience plastic deformation and as a result, the value of EA will rise. This concept is sometimes confused with tapered thin-walled structure as shown in Fig. 2.12. as studied by Nagel et al [18] and Mahmoodi et al [21]; however, the main difference is that tapered thin-walled structures has uniform wall thickness

throughout the structure because only the external appearance of the structure is tapered, whereas on a structure with functionally graded thickness the wall thickness is tapered, Fig. 2.13.

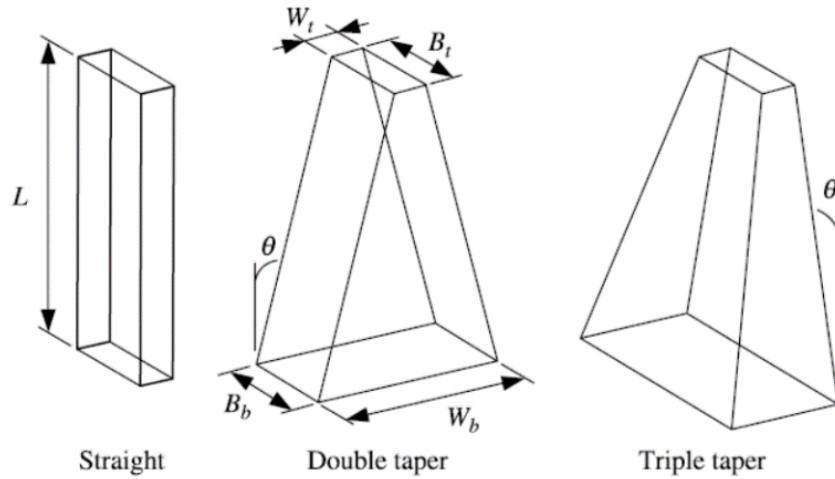


Figure 2.12 Tapered thin-walled structures. [18]

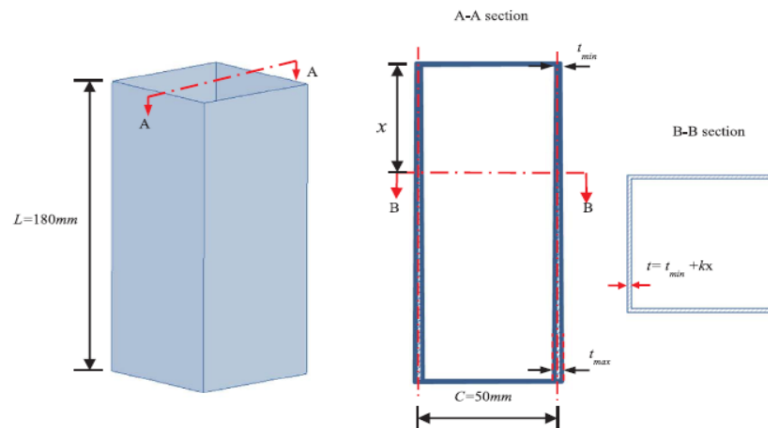


Figure 2.13 Longitudinal functionally graded thickness structure. [19]

In general, there are two directions in which taper could be applied, in the longitudinal and transverse direction, as demonstrated in Fig. 2.14. Tapering the wall thickness on the longitudinal direction lowers the peak load at the beginning of the crush because there are less materials at the tip of the structure. This type of taper is usually applied to circular tubes and frusta. Meanwhile, tapering the wall thickness in the transverse direction is done to increase the amount of material in the corners of the cross-section which undergoes severe deformation during axial loading.

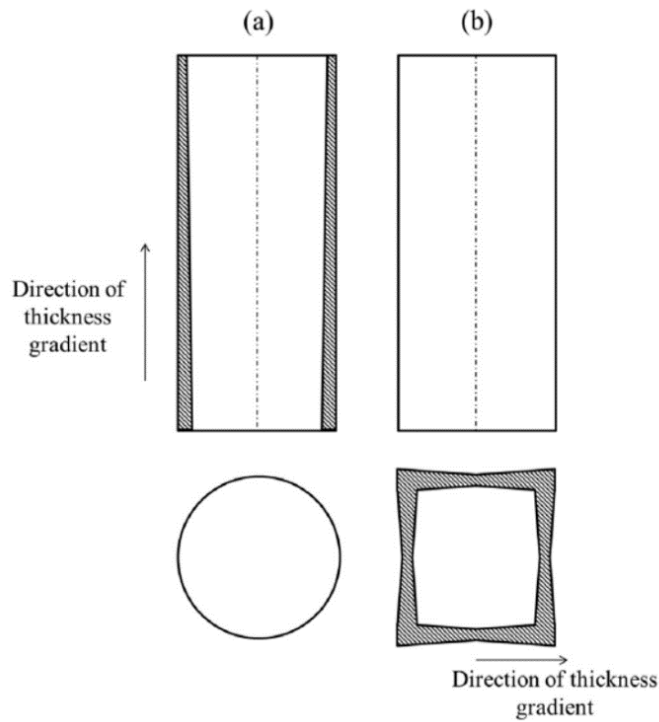


Figure 2.14 Functionally graded thickness, (a) in the longitudinal direction and (b) in the transverse direction. [3]

2.5 Transverse Functionally Graded Thickness Multi-Cell Crash Box

A transverse functionally graded thickness (TFGT) multi-cell crash box is a crash box designed to be the main energy absorber of a vehicle when subjected to axial impact loads during medium to high velocity that incorporates a combination of multi-cell and transverse functionally graded thickness concepts. Both concepts aim to increase the energy absorption capabilities of a structure although with different methods. Multi-cell structures added internal corner elements to a conventional thin-walled structure whereas transverse functionally graded thickness structures increase the amount of material on the corner element where the most severe plastic deformation occurs. A crash box incorporated with both concepts will have an appearance of a multi-cell structure but with the wall thickness tapered away from the corners, as illustrated in Fig. 2.15.

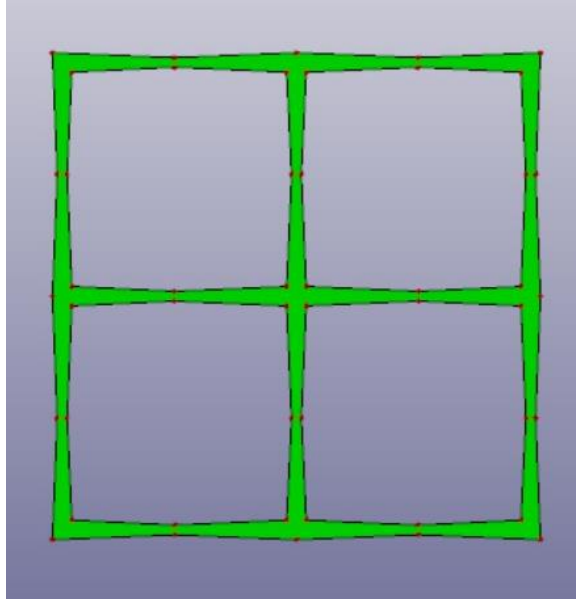


Figure 2.15 Transverse functionally graded thickness multi-cell crash box.

An important parameter for this configuration of crash box is a taper ratio, Fig. 2.16. This parameter shows the degree of taper applied to the walls which will affect the deformation mode of the crash box and the overall energy absorbing capability

$$\text{Taper Ratio} = \frac{t_c}{t_m} \quad (13)$$

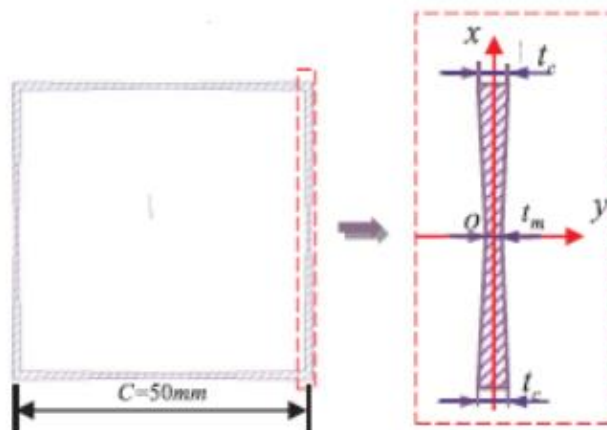


Figure 2.16 Taper ratio. [19]

On structures with transversely directed functionally graded thickness there are two types of graded thickness distribution that could be applied, which are single surface gradient (SSG) and double surface gradient (DSG) as shown in Fig. 2.17. SSG structures are tapered on only one side of the material surface, leaving the other surface completely straight whereas DSG structures are tapered on both sides. In this particular research, DSG will be applied to the crash box.

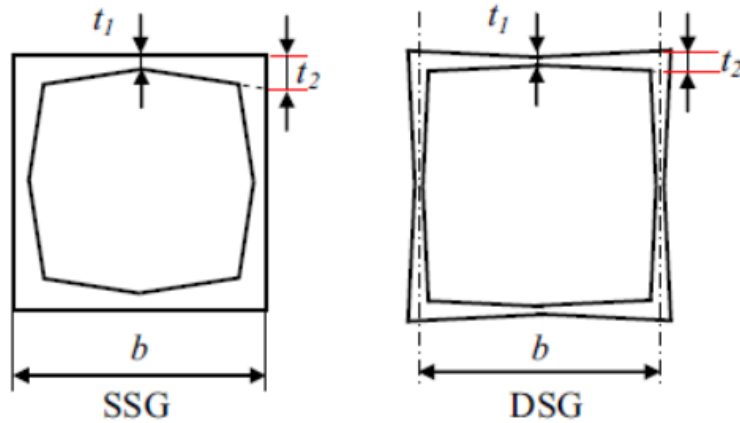


Figure 2.17 Graded thickness distribution. [20]

The theoretical calculation for mean crushing force will also be based on the super folding element method.

2.6 Trigger

Triggers are design optimizers that alters the geometrical configuration of a crash box in order for the structure to achieve favorable deformation behavior. It is typically located in the region where the load is initially introduced so that it can initiate and control the folding deformation sequence [31]. A properly designed trigger should reduce the first peak force and controls the progressive buckling without affecting the energy absorption capability of the crash box. There are various types of triggers that could be applied on a crash box, and each offers different deformation altering properties.

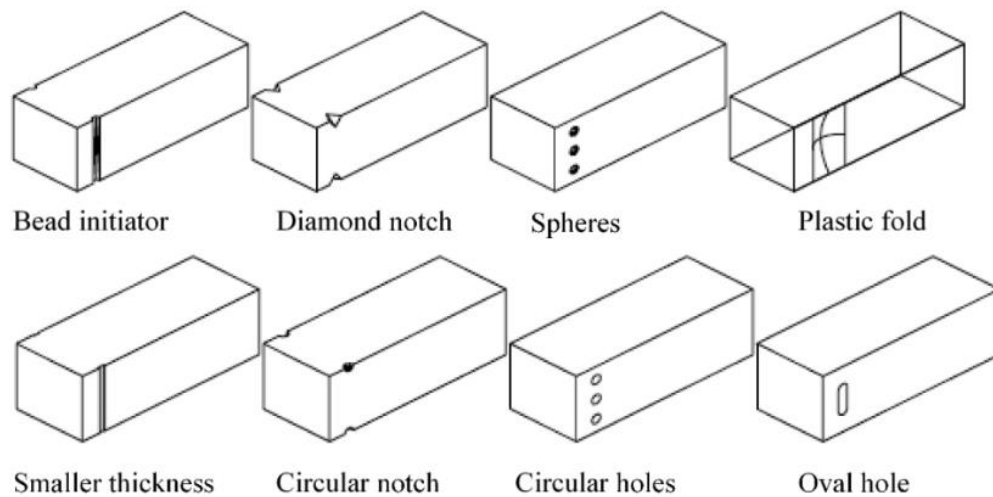


Figure 2.18 Various types of crash box triggers. [32]

Generally, there are two types of crash box triggers, stamped triggers and punched triggers. Stamped triggers introduced weaknesses in the structure of a crash box by pressing a previously determined shape into the crash box material whereas punched triggers introduced weaknesses in the structure of a crash box by removing material from a specific region within the structure. Fig. 2.18 illustrates the various types of crash box triggers with the upper row being the stamped triggers and the bottom row being the punched triggers.

According to Kumar et al [32], the bead initiator (trigger) offers the lowest first peak level and a very stable folding pattern. In addition, it also provides the most regular force curve and an expected wavelength. Due to this, a bead initiator will be applied to the transverse functionally graded thickness multi-cell crash box as a mean to improve its deformation mode and folding sequence pattern.

2.7 Finite Element Method

Finite Element Method (FEM) is a general numerical method that is developed to solve partial differential equations in two or three space variables. The approach that is done by this method is to subdivide a large system into a smaller system of simpler parts usually called finite elements. This subdividing is achieved by creating a mesh of the object made from large amounts of small elements which has a finite number of points hence the term finite element [33]. The behavior of

small finite elements when subjected to external changes are easier to predict due to it having relatively simpler equations. These simple equations are then assembled into a larger system of equations that serves as a model of the entire problem as shown in Fig. 2.19. A solution is then approximated by minimizing the error.

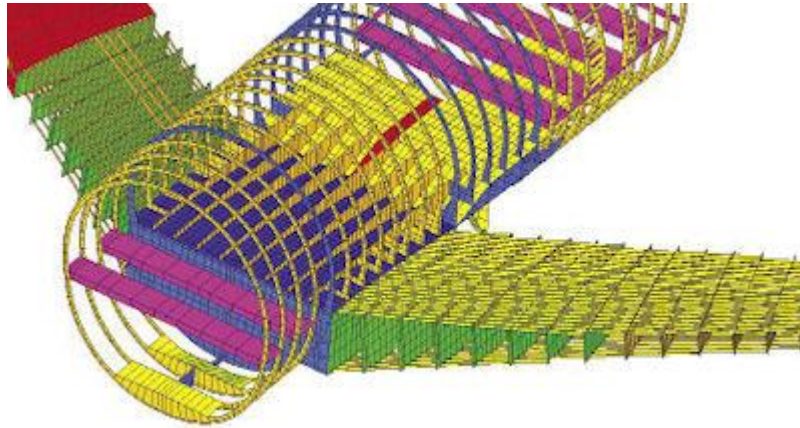


Figure 2.19 Finite element model of an aircraft. [36]

LS-DYNA is a multiphysics simulation software that offers finite element analysis using explicit time integration. It is typically used in the engineering field to simulate changes in boundary conditions, deformations, nonlinear material, impact, explosions, and manufacturing. This FEM software will be used to conduct numerical simulation in this research.

This particular research implements a solid finite element in order to properly model the functionally graded thickness characteristic of the crash-box wall structure. Solid finite elements are three dimensional elements that are able to model solid bodies and structures without any prior geometric simplification. There are several solid element formulations that could be implemented in the simulation. Element formulation 1 is underintegrated and therefore had constant stresses within the element, Fig. 2.20. It is efficient and accurate but needs hourglass stabilization. Element formulation 2 applies a selectively reduced integrated brick element which alleviates volumetric locking, Fig. 2.20. It requires no hourglass stabilization but is far slower than element formulation 1.

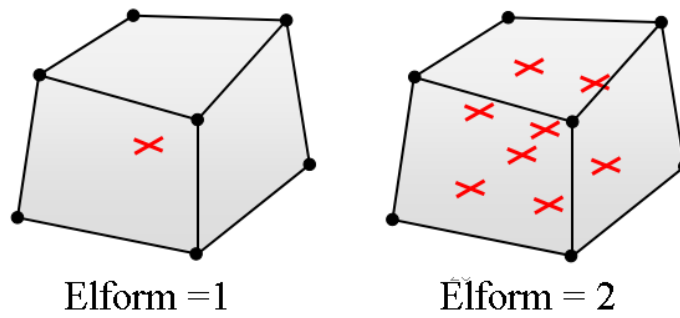


Figure 2.20 Illustration of (a) element formulation 1, (b) element formulation 2.

However, there are newer solid element formulation -1 and -2 that accounts for poor element aspect ratio. Element formulation -1 and -2 is identical with element formulation 2 but accounts for poor aspect ratio in order to reduce shear locking. The difference between both is that element formulation -1 is the more efficient formulation and element formulation -2 is the more accurate formulation. The only drawback of element formulation -1 is that it sometimes has hourglass tendencies which indicates that the result should be as accurate as element formulation -2 as long as hourglass does not occur. Element formulation -1 will be applied in the numerical simulation of this research and the hourglass energy will serve as an indicator on the accuracy of the result.

2.8 Parametric Study and Sensitivity Analysis

Parametric study is conducted to observe changes that occurred to the performance of a certain system when design variables (parameters) are varied. The conclusion of such studies should determine how a specific parameter affect the performance of a system. This kind of study is usually followed by a sensitivity analysis to determine how sensitive a parameter is to the overall performance of a system.

The sensitivity of a certain system subject to a change in design variables could be obtained by first defining the base case of the model. Next, a new output variable is calculated based on a new input variable, leaving all other parameters and assumptions unchanged. Finally, the sensitivity value could be determined by dividing the percentage change in the output variable by the percent change in the

input variable. The resulting number is dimensionless and illustrates how much the output will change when input is altered. The closer the value is to zero indicates a lower contribution to the output which means a lower sensitivity. The formula for sensitivity is shown below,

$$Sensitivity = \frac{\frac{\Delta Output}{Output_1}}{\frac{\Delta Input}{Input_1}} = \frac{\left(\frac{Output_2 - Output_1}{Output_1}\right)}{\left(\frac{Input_2 - Input_1}{Input_1}\right)} \quad (14)$$

If there are more than one sensitivity value for a system, the values could be averaged to determine the average sensitivity value within the system and boundary condition. This average value will represent the sensitivity of the parameter.

CHAPTER III

NUMERICAL MODEL

3.1 Validation Model

In the previous chapter, basic theories from literatures that serves as the foundation of this research has been defined, which allows the numerical modelling process to begin. The numerical model is first validated using experimental data obtained from previous research. The baseline model for this finite element method simulation is based on the experiment conducted by Zhang et al [20]. They conducted quasi-static axial crush experiment on 8 specimens, 4 of which is tapered on a single side and the rest tapered on both sides. The experiments were conducted on a 100 KN capacity INSTRON 5882 materials testing machine that utilizes a computer to control the input and acquire the output. The quasi-static loading is achieved by moving the top plate of the machine vertically downward towards the specimen with a displacement of 1 mm/ms.



Figure 3.1 Instron 5882.

Specifically, the validation model that is used as a baseline in this research is based on specimen with double sided taper with a taper ratio of 2 (0.8, 1.6). It is 36 mm $\times$ 36 mm and 126 mm in length with a thickness of 0.8 mm at the center and 1.6 mm at the corner, Fig 3.2. Trigger as a design optimizer is applied on the validation model at the half folding length to help initiate the folding sequence.

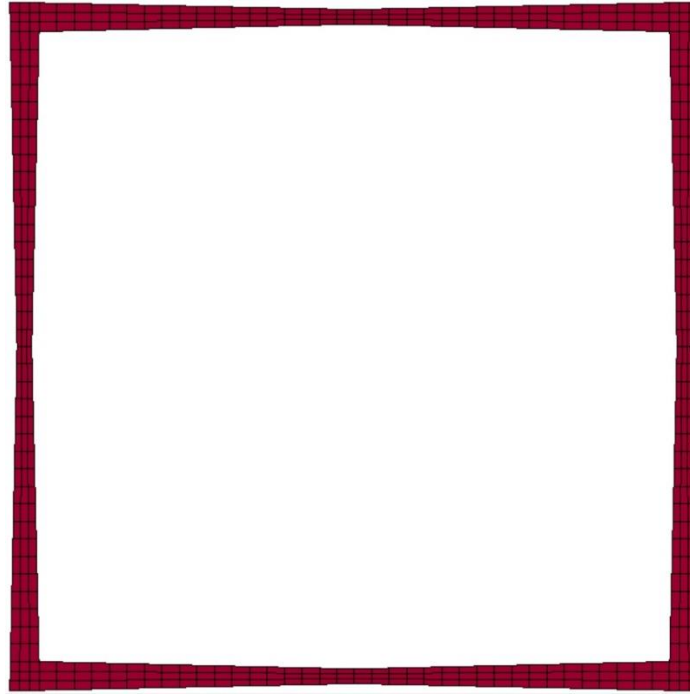


Figure 3.2 Double sided taper TFGT tube, TR=2 (0.8, 1.6).

The specimens are made using aluminum AA 6061 O and detailed properties for this material are shown in Table 3.1. Fig. 3.2 shows the stress-strain curve of AA 6061 O.

Table 3.1 AA 6061 O mechanical properties. [20]

Mechanical Property	Value
Density	2700 kg/m ³
Young's Modulus	68 GPa
Poisson Ratio	0.33
Yield Stress	71 Mpa
Strain Rate Parameter (C)	41.0133
Strain Rate Parameter (P)	6.2

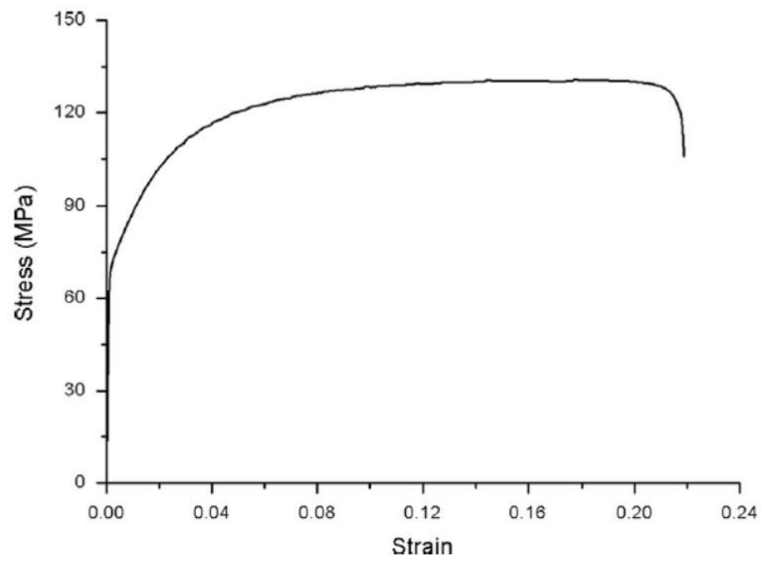


Figure 3.3 ASTM A-36 mild steel stress-strain curve. [20]

Once the baseline model geometrical configuration, experiment conditions, and material has been established, numerical modelling is done on LS-DYNA to validate the simulation parameters that will be implemented on the impact simulation of transverse functionally graded thickness multi-cell crash box.

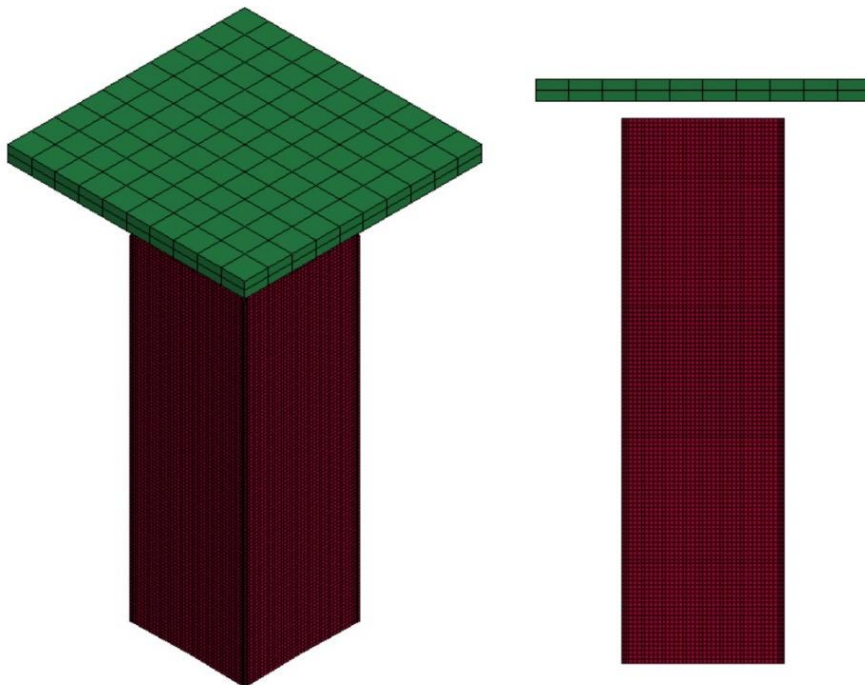


Figure 3.4 Isometric and front view of the validation model.

An Impactor is placed 4 mm above the specimen to simulate the top plate of the machine and is given a boundary so that it can only move vertically on the z axis, Fig. 3.4. Additionally, a completely fixed boundary is applied on the bottom portion of the specimen to prevent it from translating or rotating. A displacement of 1 mm/ms is given to the impactor on the negative z direction to simulate the compressive quasi-static loading.

Element formulation -1 is used on this baseline model to determine its feasibility for this research. It is previously stated in subchapter 2.7 that as long as there is no hourglass energy then the result should be as accurate as element formulation -2. For contact, automatic nodes to surface is applied between the impactor and the specimen whereas automatic single surface is applied within the specimen to prevent the elements from passing through each other. Friction between the specimen and the impactor is set to 0.4 for static and 0.3 for dynamic. Friction between the specimen material itself is set to 1.2 for static and 1.4 for dynamic.

Additional numerical simulation parameters are configured in keyword control to consider hourglass energy calculation, stonewall energy dissipation, sliding interface energy dissipation, and Rayleigh energy dissipation so that the simulation yields an accurate result. Termination is then set to 90 milliseconds and numerical modelling of the baseline model is completed.

3.2 Impact Simulation Design, Parameters, and Condition

During regular usage, vehicles generally travel forward, with reverse only being used to maneuver, meaning that during a high-speed accident the crash box structure of a vehicle will most likely experience axial loading. Due to this fact, a dynamic axial impact load will be subjected onto the crash box specimens to simulate a frontal impact. Simulation will be conducted using Finite Element Method (FEM) approach in LS-DYNA. The simulation parameters will be uniform to that of the validation model to ensure uniformity and the only difference between the validation and the simulation specimens is that the simulation specimen will be subjected to dynamic axial impact loads. There is also slight alteration in terms of the specimen cross-sectional geometry, but the meshing quality is identical.

The crash box specimens are square tubes with varying cross-sectional geometries depending on the configuration. The general dimensions of the square tubes are 100 mm $\times$ 100 mm and 400 mm in length with a 2 mm base thickness as shown in Fig. 3.5. These general dimensions are determined based on numerous papers about crash boxes particularly the works of Ciampaglia et al [23] and Sadjad et al [24].

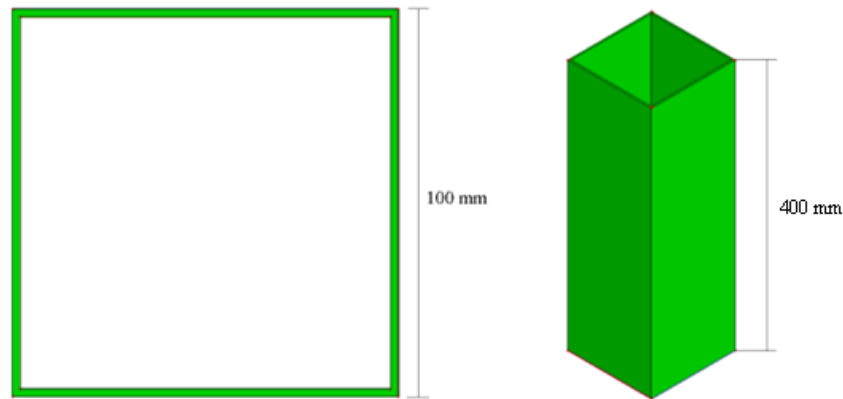


Figure 3.5 Top and isometric view of the square tube.

During the impact simulation, the crash box specimen will be oriented vertically on the z axis and the bottom portion of the specimen will be fixed. A rigid impactor is placed 5 mm above the top of the specimen and is fixed on all axis except on the z direction because the impactor will be given an initial velocity of 30 mph or 13.4 m/s on the negative z axis towards the specimen. With an impactor mass of 98 kg the total kinetic energy of the impactor should be around 8.798 kJ. Figure 3.6 illustrates the specimen and the impactor initial position during simulation. The combined simulation parameters are shown on table 3.2.

Table 3.2 Combined simulation parameters.

Parameter	Value
Loading Type	Dynamic Axial
Impactor Type	Rigid Body
Impactor Velocity	30 mph or 13.4 m/s
Impactor Weight	98 kg
Total Kinetic Energy	8.798 kJ

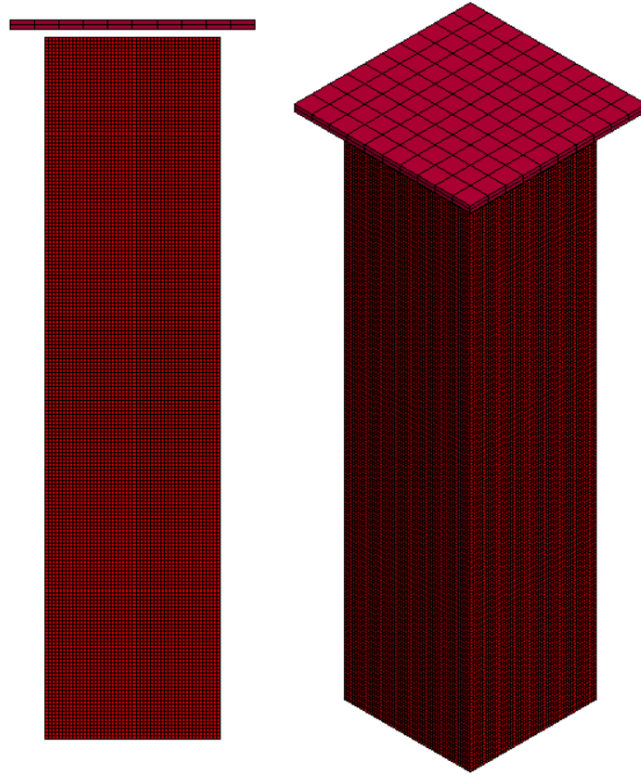


Figure 3.6 Initial position of the simulation.

3.3 Specimen and Impactor Geometries

As stated on section 3.1 the general dimensions for the crash box specimens are $100\text{ mm} \times 100\text{ mm}$ and 400 mm in length with a 2 mm base thickness. The 2 mm thickness is only a base thickness because on some crash box configurations the thickness varied depending on the location. In total, there are 13 specimens that will be simulated using the same predetermined parameters in order to compare the specific energy absorption and crashworthiness performance. The first three are conventional geometries that will be used as a benchmark for the transverse functionally graded thickness multi-cell crash box, the next five are different taper ratio configurations of the TFGT multi-cell crash box, and the last five are different taper ratio configurations of the TFGT multi-cell crash box with design optimizer (bead trigger) applied. The conventional crash box geometries are plain square tube, 4-cell multi-cell, and TFGT. The TFGT multi-cell crash boxes are varied with taper ratios of 1.5, 1.727, 2, 2.333, and 2.75.

CAD modelling and meshing of the geometries are conducted using LS-DYNA because the geometries are relatively simple. To create the CAD, several points are established as the outline of the cross-sectional geometry and then a line curve connects the points to create the cross-section. This cross-section is then extruded as a solid to create the 3D solid of the crash-box specimen.

3.3.1 Conventional Geometries

The plain square tube crash box is 100 mm $\times$ 100 mm and 400 mm in length with a 2 mm wall thickness, Fig. 3.7. The multi-cell crash box incorporates a 4-cell design and its dimensions are 100 mm $\times$ 100 mm and 400 mm in length with a 2 mm wall thickness, Fig. 3.8. The transverse functionally graded thickness crash box shown in Fig. 3.9 is 100 mm $\times$ 100 mm in cross-section and 400 mm in length with a 2 mm corner thickness. It has a taper ratio of 2 which means that the wall has a thickness of 1 mm in the middle.

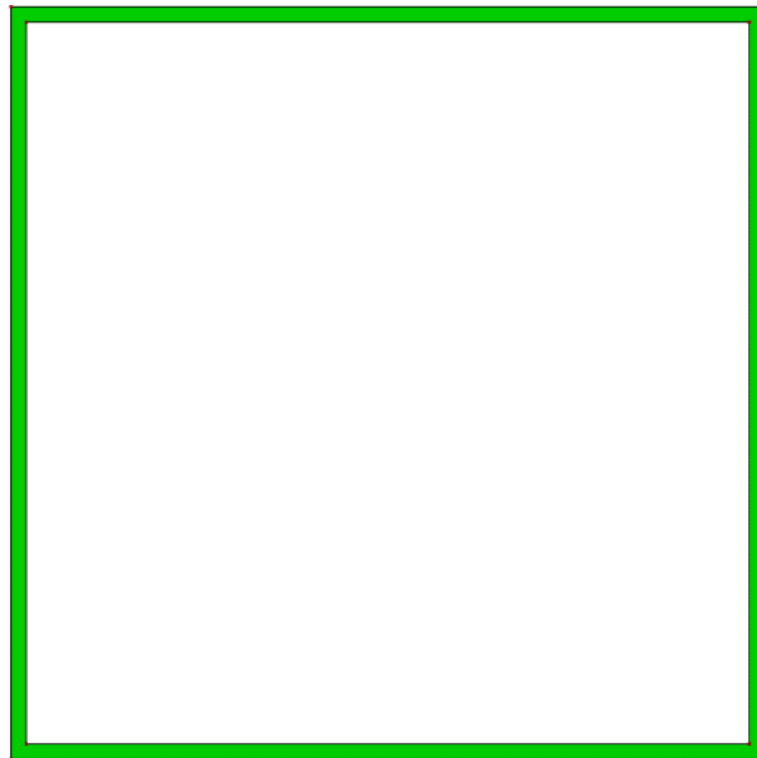


Figure 3.7 Conventional square tube crash box.

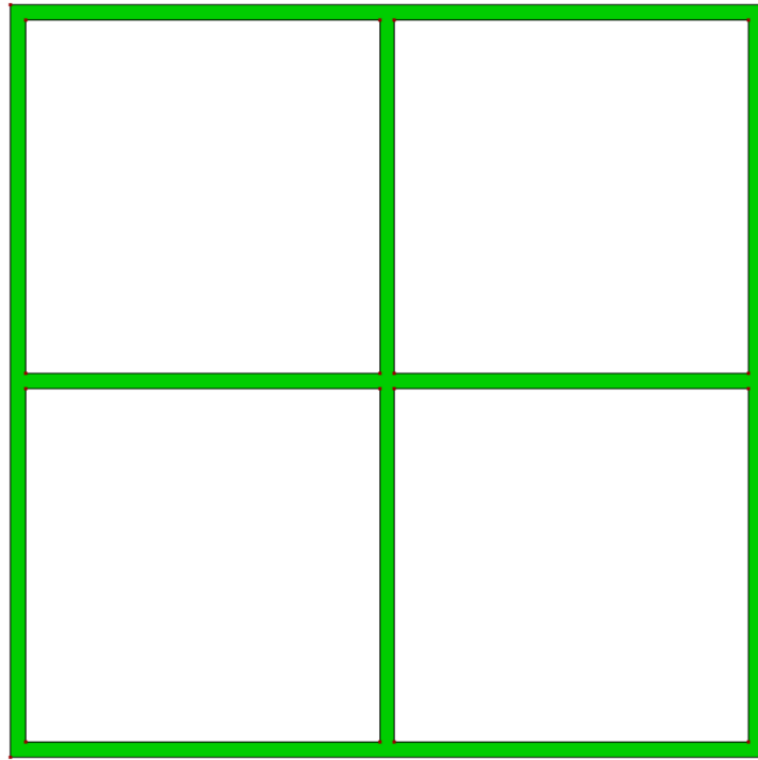


Figure 3.8 4-cell multi-cell crash box.

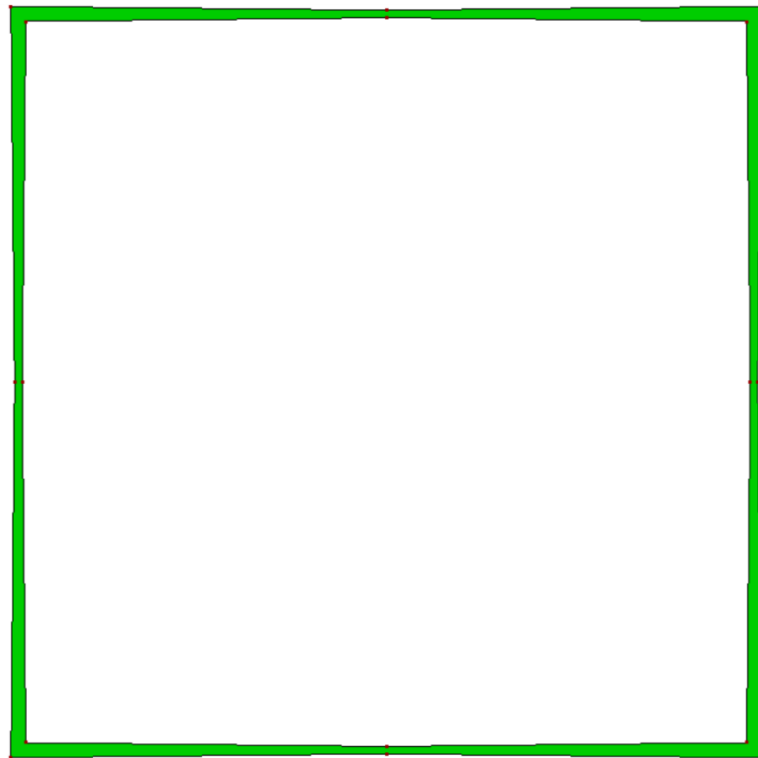


Figure 3.9 Transverse functionally graded thickness crash box.

3.3.2 Transverse Functionally Graded Thickness Multi-Cell Geometries

The baseline model for the transverse functionally graded thickness multi-cell crash box is 100 mm $\times$ 100 mm and 400 mm in length with a 2 mm base wall thickness and will be referred as TFGT multi-cell geometry c. It has a taper ratio of 2 which means that the taper has a highest thickness value of 2 mm and a smallest thickness value of 2 mm. This geometrical design also incorporates a 4-cell multi-cell cross-section. Due to this, the 1 mm wall thickness is present on 12 different points across the cross-section unlike the conventional transverse functionally graded thickness crash box shown on Fig. 3.9 that only has 4. The geometrical cross-section for the transverse functionally graded thickness multi-cell crash box with a taper ratio of 2 is shown on Fig. 3.10.

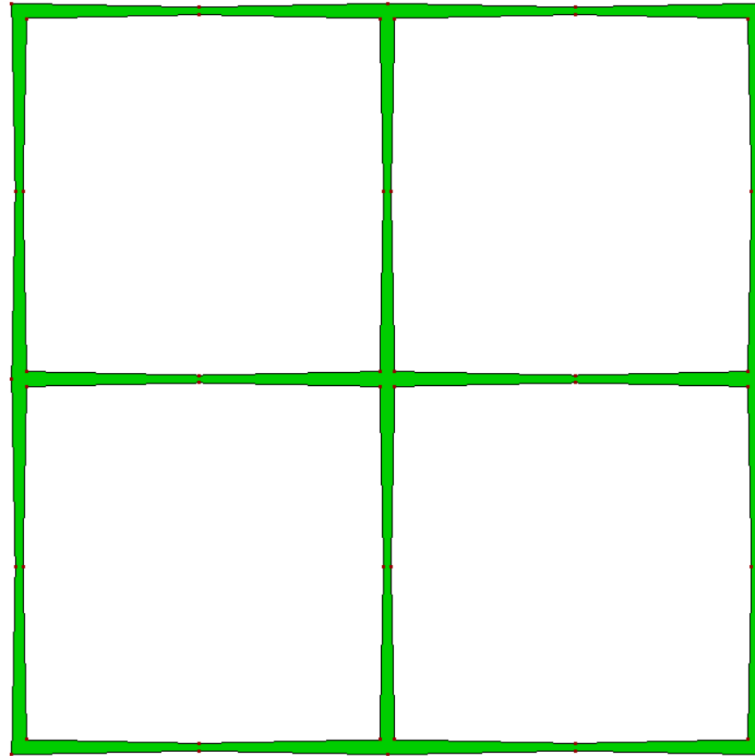


Figure 3.10 TFGT multi-cell crash box with a taper ratio of 2.

To evaluate the effect of different taper ratios on the TFGT multi-cell crash box's crashworthiness performance, four other TFGT multi-cell crash boxes with different taper ratios are also modelled.

The first geometry (TFGT multi-cell geometry a) has a maximum thickness of 1.8 and a minimum thickness of 1.2 with a taper ratio of 1.5. The second geometry (TFGT multi-cell geometry b) has a maximum thickness of 1.9 and a minimum thickness of 1.1 with a taper ratio of 1.727. The third geometry (TFGT multi-cell geometry d) has a maximum thickness of 2.1 and a minimum thickness of 0.9 with a taper ratio of 2.333. The fourth geometry (TFGT multi-cell geometry e) has a maximum thickness of 2.2 and a minimum thickness of 0.8 with a taper ratio of 2.75. The geometrical parameters are combined on Table 3.3 and the cross-section geometries (except the baseline TFGT multi-cell geometry c) are illustrated in Fig. 3.11.

Table 3.3 Geometrical parameters for TFGT multi-cell crash boxes.

Geometry	Max. Thickness	Min. Thickness	Taper Ratio (TR)
A	1.8	1.2	1.5
B	1.9	1.1	1.727
C	2	1	2
D	2.1	0.9	2.333
E	2.2	0.8	2.75

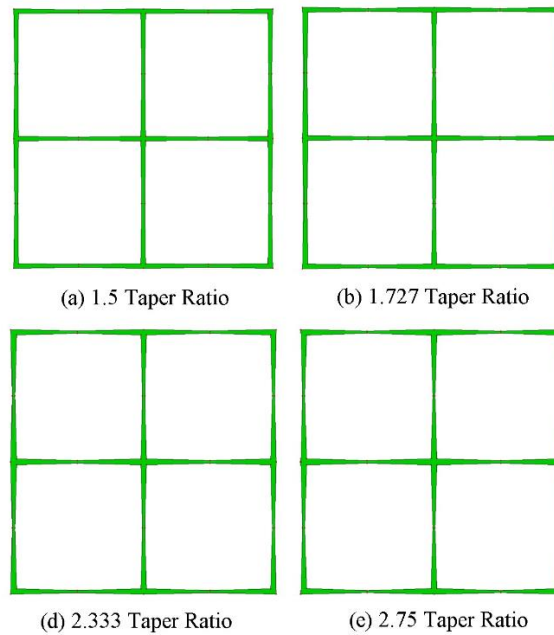


Figure 3.11 TFGT multi-cell cross-sectional geometries.

3.3.3 Impactor Geometry

The impactor is 140 mm $\times$ 140 mm and 5 mm in height. It is set into a rigid body in order to simulate crashing into relatively sturdier structures such as a wall or concrete road barriers and is situated 5 mm above the top of the crash box specimen with an initial velocity of 30 mph or 13.4 m/s in the negative z direction. The dimensions are slightly larger than the cross-section of the crash box specimens which is only 100 mm $\times$ 100 mm because it needs to accommodate material deformation on the x and z axis. The geometry is modelled using the LS-DYNA shape mesher as a box solid. There are 10 elements on each side (x and y axis) and 2 elements height wise (z axis) with a combined total of 200 elements. The impactor geometry is shown on Fig. 3.12 and Fig. 3.13.

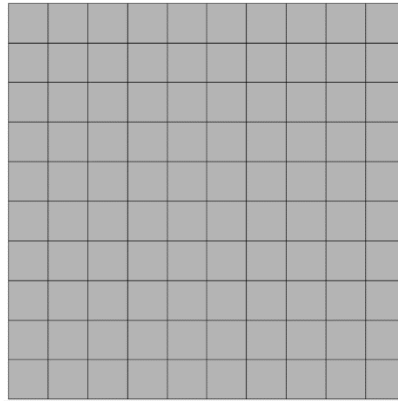


Figure 3.12 Rigid body impactor top view.

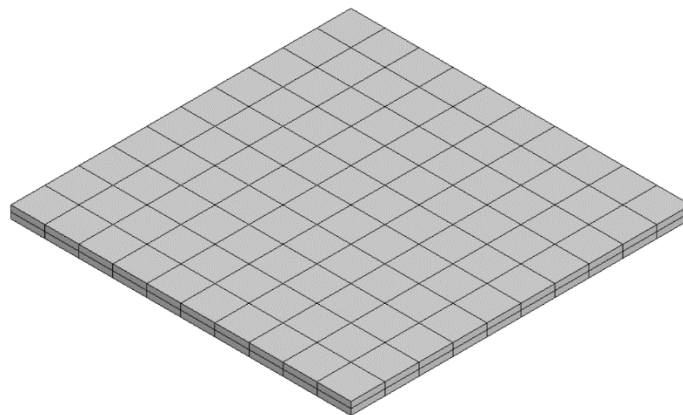


Figure 3.13 Rigid body impactor iso view.

3.3.4 Meshing

Meshing is conducted to adapt the geometrical shape of the crash box CAD model into smaller finite elements. For the numerical model of this research, meshing is done using Altair HyperMesh to introduce three solid elements on the thickness of the thin-walled tube. Three elements are placed on the thickness based on previous mesh convergence analysis conducted by Zhang et al [20] which concludes that three elements are enough for solid elements to achieve an accurate result. The finite element size is dynamic which adjusts according to the thickness of the wall. The largest element is a third of the maximum thickness and the smallest element is a third of the minimum thickness. Fig. 3.14 until Fig. 3.21 shows the meshing quality of the non-triggered crash boxes.

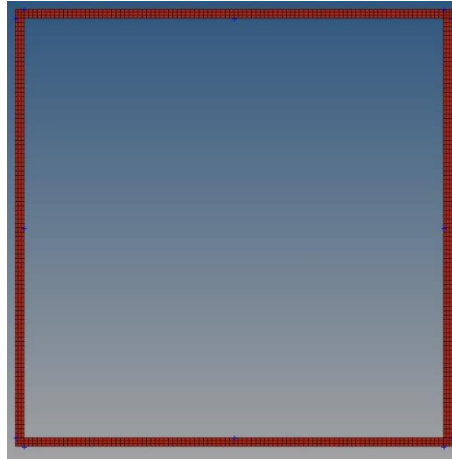


Figure 3.14 Square crash box mesh.

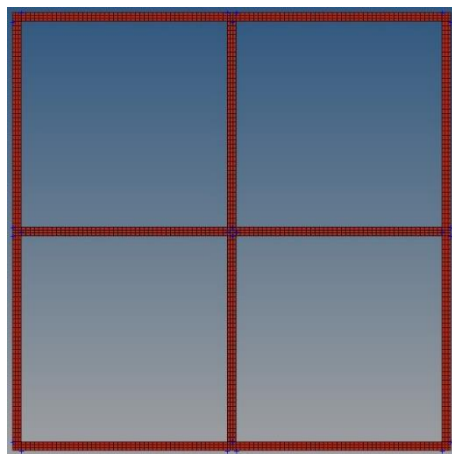


Figure 3.15 4-cell multi-cell crash box mesh.

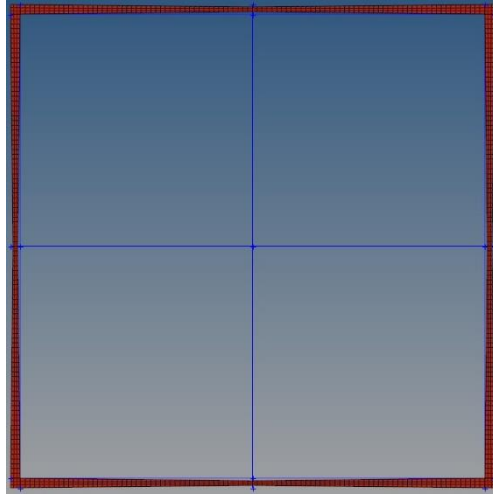


Figure 3.16 Transverse functionally graded thickness crash box mesh.

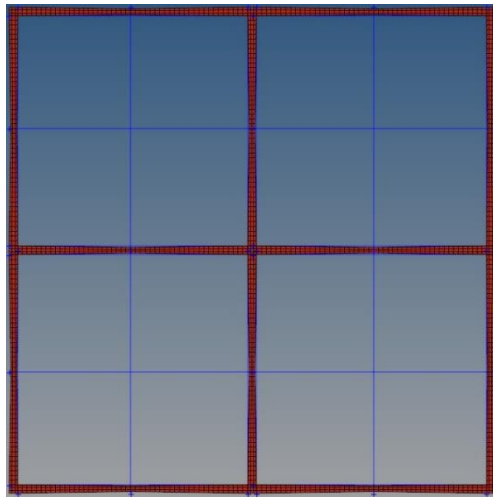


Figure 3.17 TFGT multi-cell crash box A (TR=1.5) mesh.

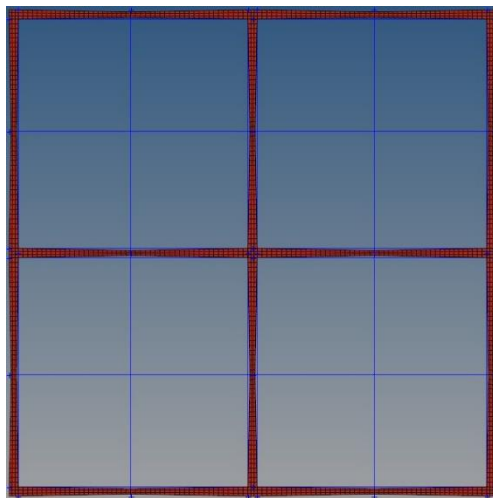


Figure 3.18 TFGT multi-cell crash box B (TR=1.727) mesh.

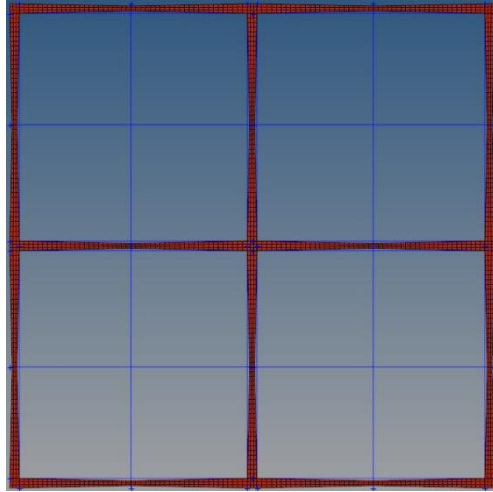


Figure 3.19 TFGT multi-cell crash box C (TR=2) mesh.

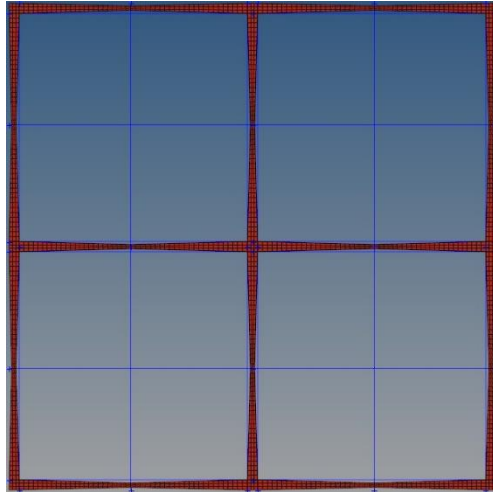


Figure 3.20 TFGT multi-cell crash box D (TR=2.333) mesh.

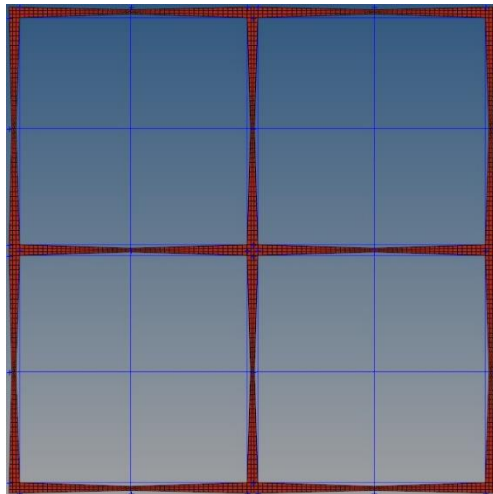


Figure 3.21 TFGT multi-cell crash box E (TR=2.75) mesh.

3.4 Trigger Application

Based on the research of Kumar et al [32], design optimizer in the form of bead triggers will be applied on all specimens. Due to limitations in the finite element geometry, the shape of the bead is more similar to a triangle than a half circle. However, since the main purpose of a trigger is to introduce weaknesses within the structure of a crash box, the triangular bead trigger should still induce a favorable deformation behavior. In addition, Harhash et al [31] also applied a triangular bead trigger on their research and it yielded positive results.

The trigger is placed on the front and the back of the crash boxes and is positioned on its respective half folding length. The folding length for each specimen is obtained from initial simulations, Table 3.4. The depth of the trigger is determined to be quarter the thickness for non-graded specimens and align with the minimal thickness for the graded specimens.

Table 3.4 Half folding lengths of TFGT multi-cell crash boxes.

Crash Box	Half Folding Length (mm)
Square tube	41
Multi-cell tube	24
TFGT tube	26
TFGT multi-cell A (TR=1.5)	25
TFGT multi-cell B (TR=1.727)	21
TFGT multi-cell C (TR=2)	18
TFGT multi-cell D (TR=2.333)	17
TFGT multi-cell E (TR=2.75)	16

In the numerical model, triangular bead triggers are introduced on the crash box geometry by editing the node so that it aligns with the trigger depth. The crash box geometries with trigger are shown on Fig. 3.22 until Fig. 3.29. It is noticeable from the figures that with increasing taper ratio, the value half folding length decreases and the trigger position became higher.

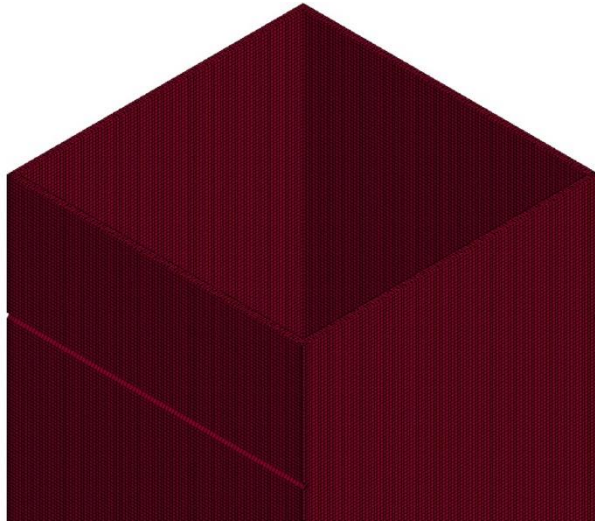


Figure 3.22 Square tube with trigger.

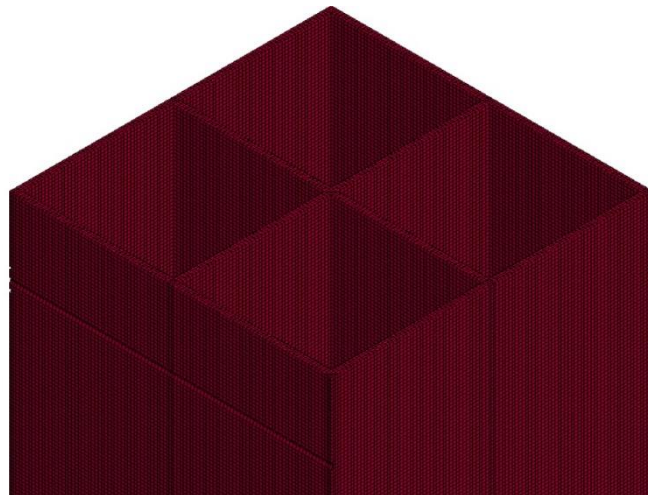


Figure 3.23 Multi-cell tube with trigger.

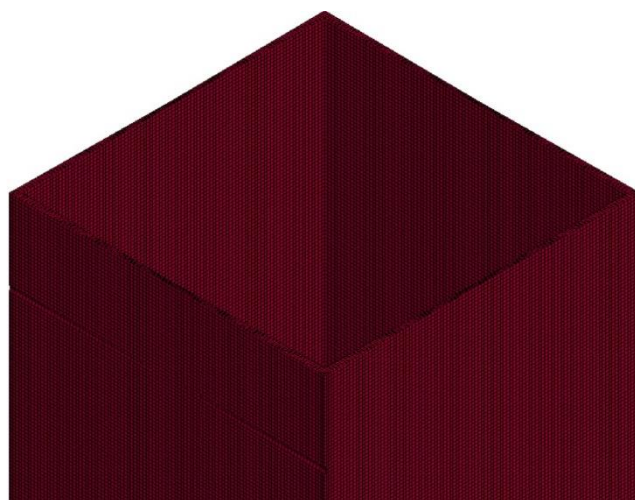


Figure 3.24 TFGT tube with trigger.

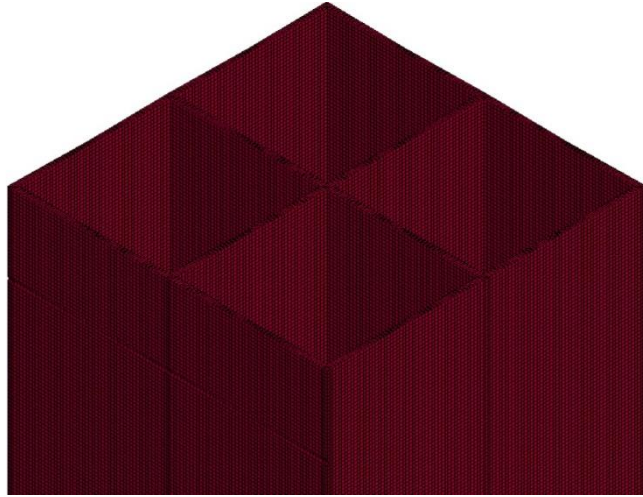


Figure 3.25 TFGT multi-cell A ($TR=1.5$) with trigger.

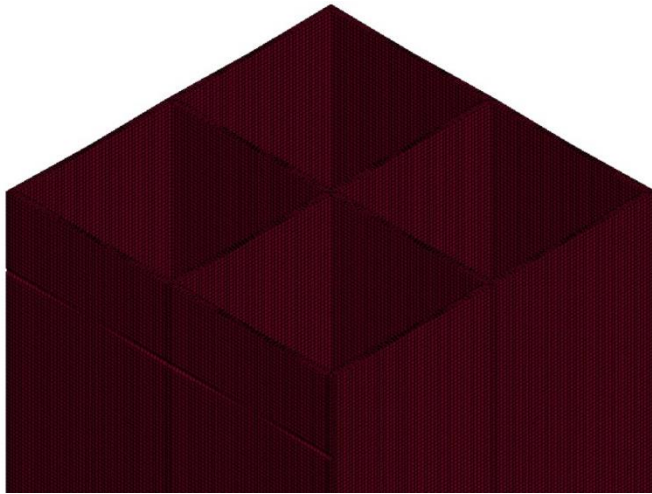


Figure 3.26 TFGT multi-cell B ($TR=1.727$) with trigger.

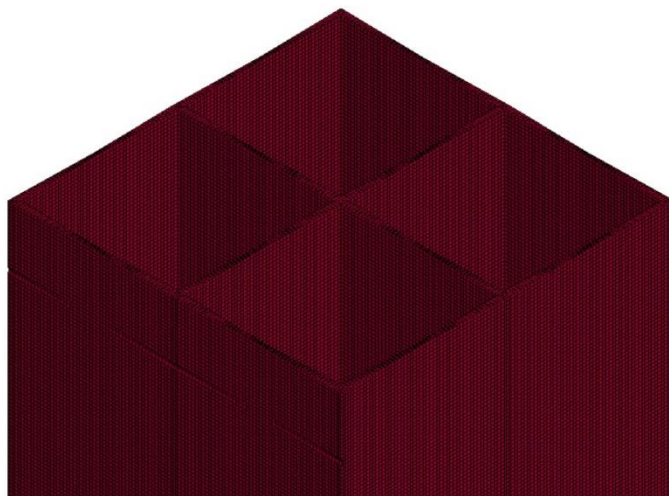


Figure 3.27 TFGT multi-cell C ($TR=2$) with trigger.

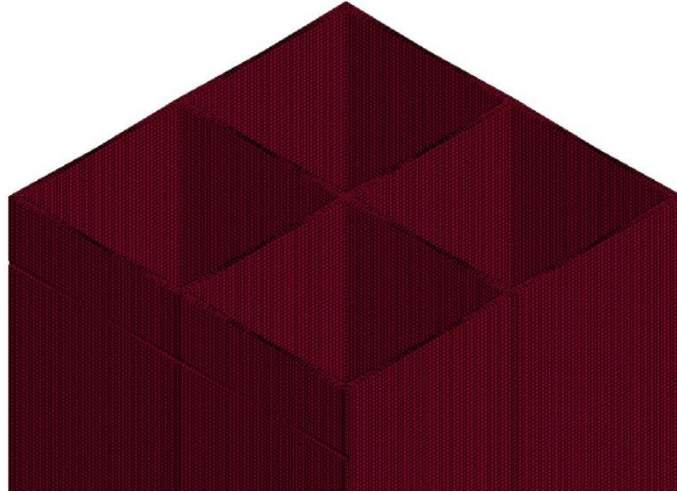


Figure 3.28 TFGT multi-cell D (TR=2.333) with trigger.

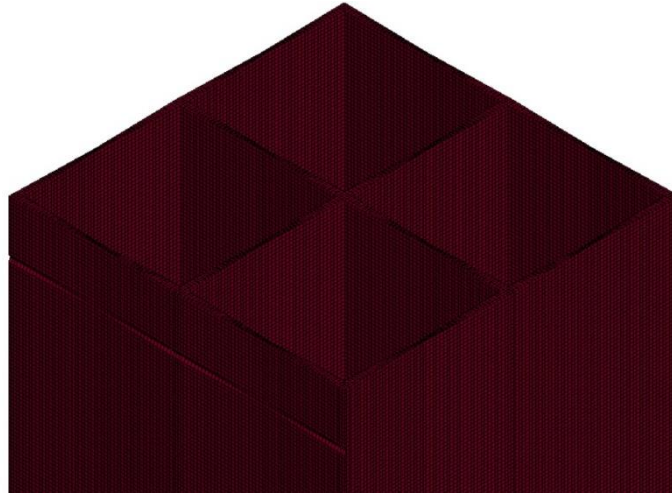


Figure 3.29 TFGT multi-cell E (TR=2.75) with trigger.

3.5 Material

Aluminum 6060-T4 is uniformly used on all crash box specimens. This type of aluminum is a 6000 series aluminum alloy in the wrought aluminum-magnesium-silicon family [25]. It is closely related to aluminum alloy 6063 with the main difference being its magnesium content which is slightly lower in aluminum alloy 6060. This type of aluminum alloy is usually heat treated in order to produce tempers with a higher strength and lower ductility. Aluminum alloy 6060-T4 has the following composition, shown on Table 3.5. [26].

Table 3.5 Aluminum 6060-T4 composition.

Component	Composition (%)
Aluminum (Al)	97.9 – 99.3
Magnesium (Mg)	0.35 – 0.6
Silicon (Si)	0.3 – 0.6
Iron (Fe)	0.1 – 0.3
Zinc (Zn)	0 – 0.15
Manganese (Mn)	0 – 0.1
Titanium (Ti)	0 – 0.1
Copper (Cu)	0 – 0.1
Chromium (Cr)	0 – 0.05
Residuals	0 – 0.15

The T in the aluminum 6060-T4 stand for heat treated alloys and depending on the numerical designation after the T, the basic type of the heat treatment will differ. T4 indicates that the alloy has been given a solution heat treatment and without any cold work, it is naturally aged to a stable condition. The mechanical properties of Aluminum 6060-T4 is defined on Table 3.6 and is obtained from a combination of reference [26], [27], and [28].

Table 3.6 Aluminum 6060-T4 mechanical properties.

Mechanical Property	Value
Density	2.71 g/cm ³
Young's Modulus	68.2 GPa
Poisson Ratio	0.3
Yield Stress	71 Mpa
Strain Rate Parameter (C)	41.0133
Strain Rate Parameter (P)	6.2

Material definition for the crash box specimens in LS-DYNA utilizes MAT_PIECEWISE_LINEAR_PLASTICITY which requires a stress-strain curve. It illustrates the relationship between the material's stress and strain which is used by the software to further define the mechanical properties of the material and how it would behave in a FEM simulation. The stress strain data is shown in Table 3.7 and the stress-strain curve is illustrated in Fig. 3.30.

Table 3.7 Aluminum 6060-T4 stress-strain data.

Plastic Stress (GPa)	Strain
0.08	0
0.115	0.024
0.139	0.049
0.15	0.074
0.158	0.099
0.167	0.124
0.171	0.149
0.173	0.174

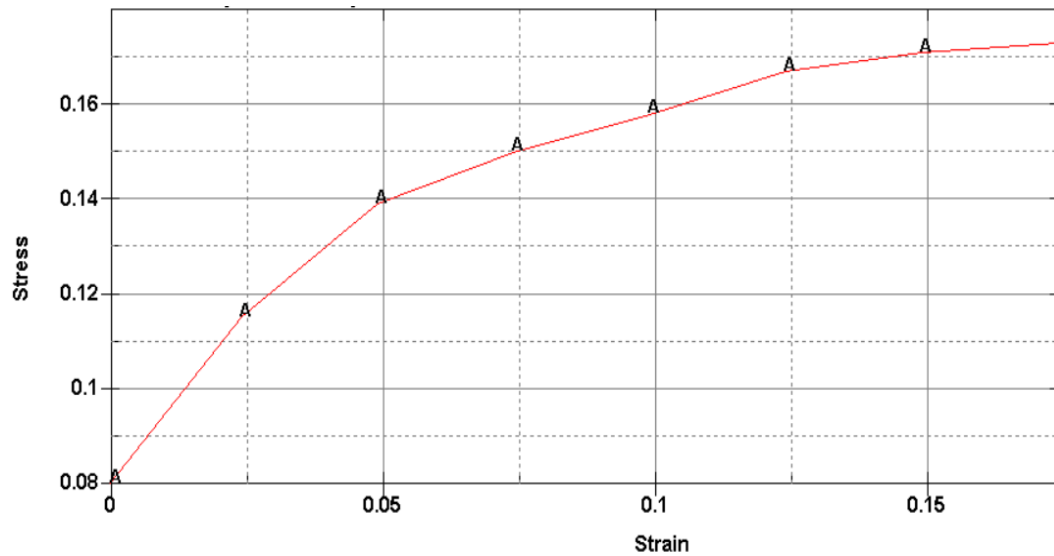


Figure 3.30 Aluminum 6060-T4 stress-strain curve.

Material definition for the impactor utilizes MAT_RIGID because the impactor is a rigid body to simulate crashing into relatively sturdier structures such as a wall or concrete road barriers. The density of the impactor is set to 0.001 kg/mm³ and is not based on any material but rather from initial simulations. When trying various densities for the impactor, initial simulations shows that the kinetic energy generated by the impactor with the aforementioned density and 30 mph or 13.4 m/s of velocity results in acceptable and logical deformation with a reasonable 15 milliseconds of simulation time control. Although this may seem unorthodox, this density definition method stays within the boundaries of the scope as long as the exact same impactor model is applied to every specimen because the main goal of this simulation is to compare the crashworthiness performance of varying cross-sectional geometries of a crash box. The properties of the impactor are shown in Table 3.8.

Table 3.8 Impactor properties.

Property	Value
Density	0.001 kg/mm ³
Mass	98 kg
Young's Modulus	250 GPa
Poisson Ratio	0.3
Initial Velocity	30 mph or 13.4 m/s (negative z direction)
Kinetic Energy	8.798 kJ

3.6 LS-DYNA Keyword Modelling

Although the base parameters for the simulation have been stated on previous sections, there are several additional parameters that needs to be defined using keyword in order to ensure that the specimen material behaves as intended, the output contains the necessary data for analysis, and the simulation ran as accurate as possible without sacrificing too much computing resources.

- SECTION

The section keyword defines what type of element the FEM parts is. For this specific application, both the impactor and specimen are solid so the SECTION_SOLID option is utilized. This keyword also allows the definition of element formulation which is crucial in determining how the material deformation is calculated. For conventional square tube and multi-cell crash boxes, element formulation 2 (designation for fully integrated S/R solid) is selected because the elements on both specimens are perfect squares. For the rest of the crash box geometries element formulation -1 (designation for fully integrated S/R solid intended for elements with poor aspect ratio, efficient formulation) is used because the cross-sectional geometry utilizes transverse functionally graded thickness which causes the elements to have a poor aspect ratio. Efficient formulation is used because it significantly reduced the simulation time for these geometries as long as hourglass did not occur.

- PART

The part keyword assigns the material and section previously defined to the finite element parts. This is done by selecting the appropriate section id and material id to the impactor and specimen.

- CONTACT

Contact is one of the most important parameters that needs to be defined in order for the impact simulation to behave realistically. Without contact, the finite elements will not interact with each other and there will be no energy transfer between the impactor and the specimen. There are two contacts that needs to be defined. First, contact between the impactor and the specimen is defined using the contact keyword option CONTACT_AUTOMATIC_NODES_TO_SURFACE with 0.4 static coefficient of friction and 0.3 dynamic coefficient of friction [29]. For this contact, the slave part is the specimen and the master part is the impactor. The second contact is defined for the specimen in order to

prevent the deformed specimen material to pass through itself. The keyword used is CONTACT_AUTOMATIC_SINGLE_SURFACE with 1.2 static coefficient of friction and 1.4 dynamic coefficient of friction. For this contact, the slave part is the specimen and there is no master part.

- CONTROL

The control keyword sets the boundaries for the simulation to determine how the simulation is ran. There are four control parameters that needs to be defined. The first control is CONTROL_ACCURACY where OSU (Objective Stress Update) is set to 1 to increase the accuracy of the impact simulation. The second control is CONTROL_ENERGY where HGEN (hourglass energy calculation option), RWEN (stonewall energy dissipation option), SLNTEN (sliding interface energy dissipation option), and RYLEN (Rayleigh energy dissipation option) is set to 2 so that sliding interface energy dissipation is computed and included in the energy balance. The third control is CONTROL_TERMINATION which is set to 15 ms. The last control is CONTROL_TIMESTEP which is set to default with a scale factor of 0.9 because this timestep scale factor value is accurate enough.

- DATABASE

The database keyword is used to determine what information is tracked during the simulation and obtained as an output when the simulation is finished. In DATABASE_ASCII_option, GLSTAT (global statistics), MATSUM (material energies), RBDOUT (rigid body data), and RCFORC (resultant interface forces) is checked with a time interval of 0.05 milliseconds. BINARY_D3PLOT is also defined with a time interval of 0.05 milliseconds.

Once all the required keyword parameters have been defined, the model is checked using the built in LS-DYNA model checking. If there are no errors in the model then the completed impact simulation model is simulated using LS-DYNA SMP double precision solver in LS-DYNA program manager. Simulation time took between 30 to 60 hours depending on the geometry.

CHAPTER IV

RESULT AND ANALYSIS

4.1 Validation Model

The validation model is simulated using LS-DYNA with double precision solver which took around 36 hours to finish. The result is accessed using LS-Prepost software and the following deformation is obtained for the numerical simulation of the validation model when subjected to quasi-static loading of 1 mm/ms, Fig. 4.1.



Figure 4.1 Validation model deformation.

The deformation mode of the validation model is inextensional and it started at the top region of the crash box which indicates that the trigger worked properly to encourage the tube to deform progressively in the desired manner. It is also observable that the contact configuration works properly due to the fact that there are no elements that penetrates through each other. With a simulation termination time of 90 milliseconds, the validation model has a maximum deformation of 90 mm and a total of three folds are observed.

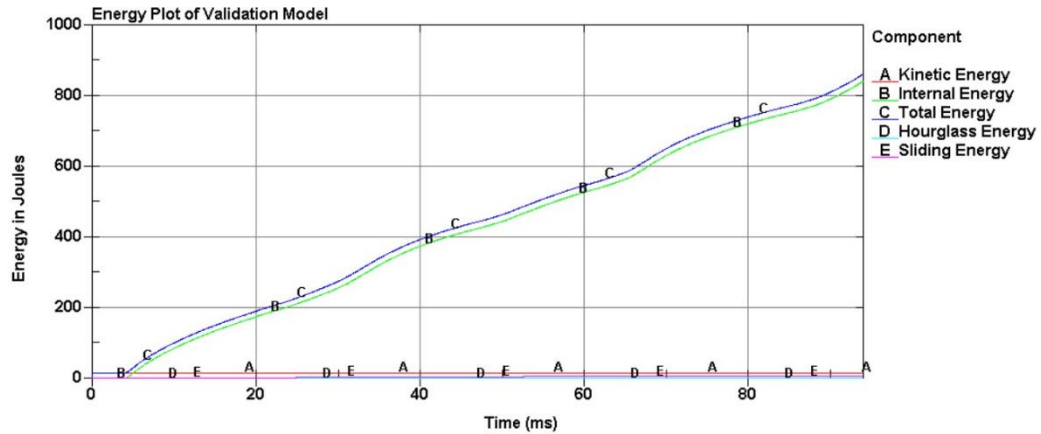


Figure 4.2 Energy plot of the baseline model.

Fig. 4.2 shows that the hourglass energy stays constantly at zero which proves that the element formulation -1 worked properly and that the result is as accurate as element formulation -2 because hourglassing does not occur. Some of the energy is dissipated as sliding energy which is why the internal energy never reached the total energy. However, this is a good thing since positive sliding energy further indicates that the contact is working properly. The force-displacement curve of the baseline model is illustrated below in Fig. 4.3.

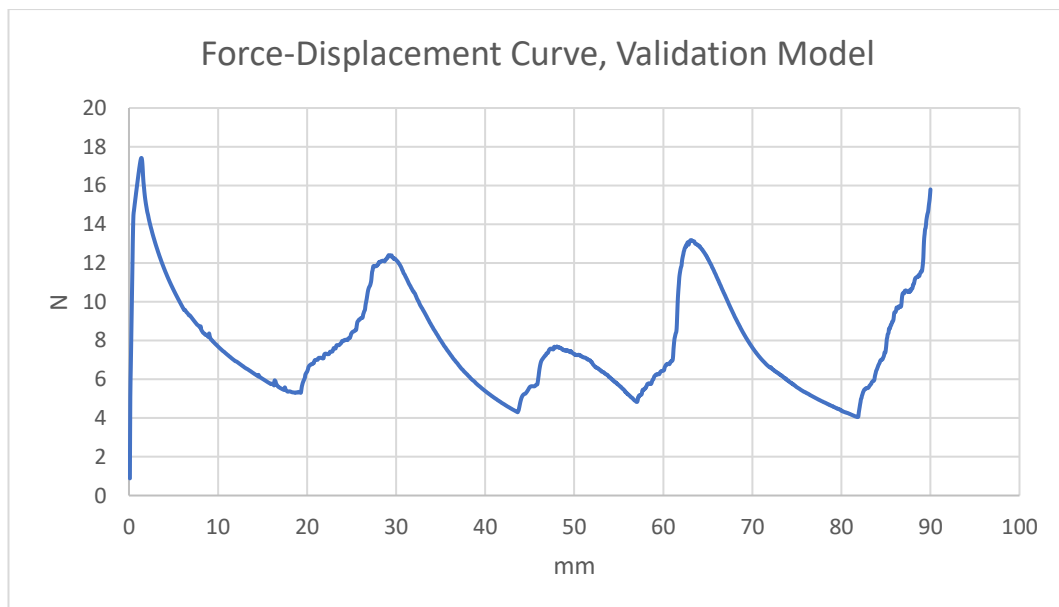


Figure 4.3 Force-displacement curve, baseline model.

Table 4.1 Crashworthiness indicators of baseline model.

Cross-sectional Area (mm ²)	172.8
Total Specimen Mass (g)	58.787
Max Displacement (mm)	90
Peak Force (N)	17.424
Energy Absorbed (J)	664.988
Specific Energy Absorption (kJ/kg)	11.312
Mean Crushing Force (N)	7.389
Crushing Force Efficiency	0.424

From the force-displacement curve, the crashworthiness performance of the baseline model could be determined by analyzing the crashworthiness indicators which are energy absorbed, specific energy absorption, mean crushing force, and crushing force efficiency, Table 4.1. Referring to the research conducted on specimen DSG (0.8, 1.6) by Zhang et al [20], the following results could be obtained.

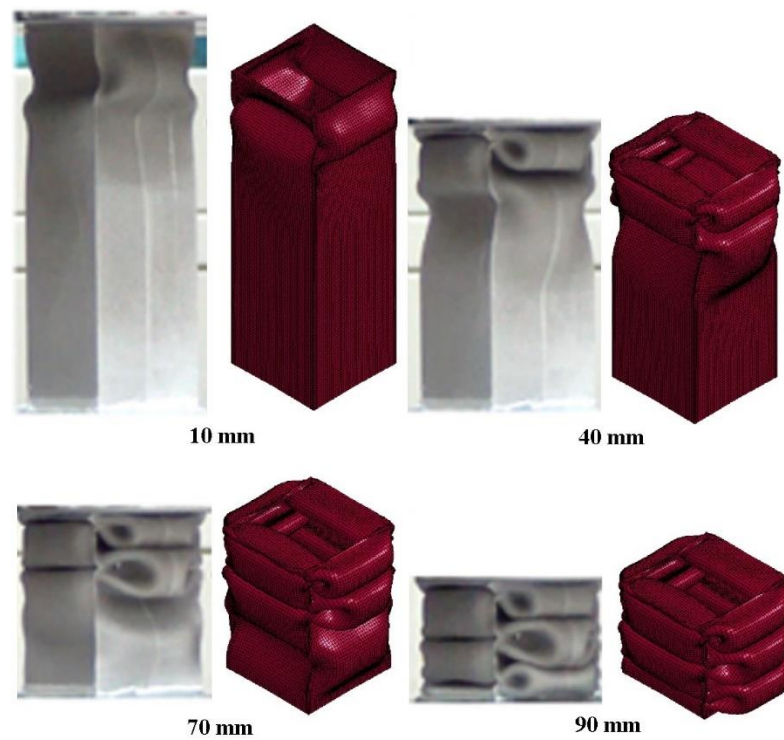


Figure 4.4 Deformation comparison.

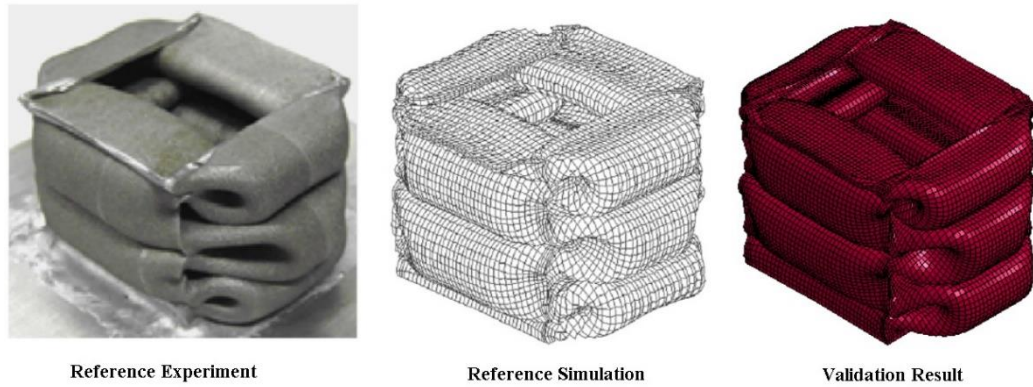


Figure 4.5 Final deformation state.

Results shows that within the first 10 mm both the experiment and the validation model started to form its first fold, Fig 4.4. When the displacement reaches 40 mm the first fold is completed and the second fold started to appear. On 70 mm displacement the second fold is formed and the third fold is initiated. The quasi-static loading is finished when the displacement reaches 90 mm and the tube is in its final deformation state with a total of three folds, Fig 4.5. Deformation on both the experiment and validation model exhibits very similar behavior which supports the accuracy of the validation model.

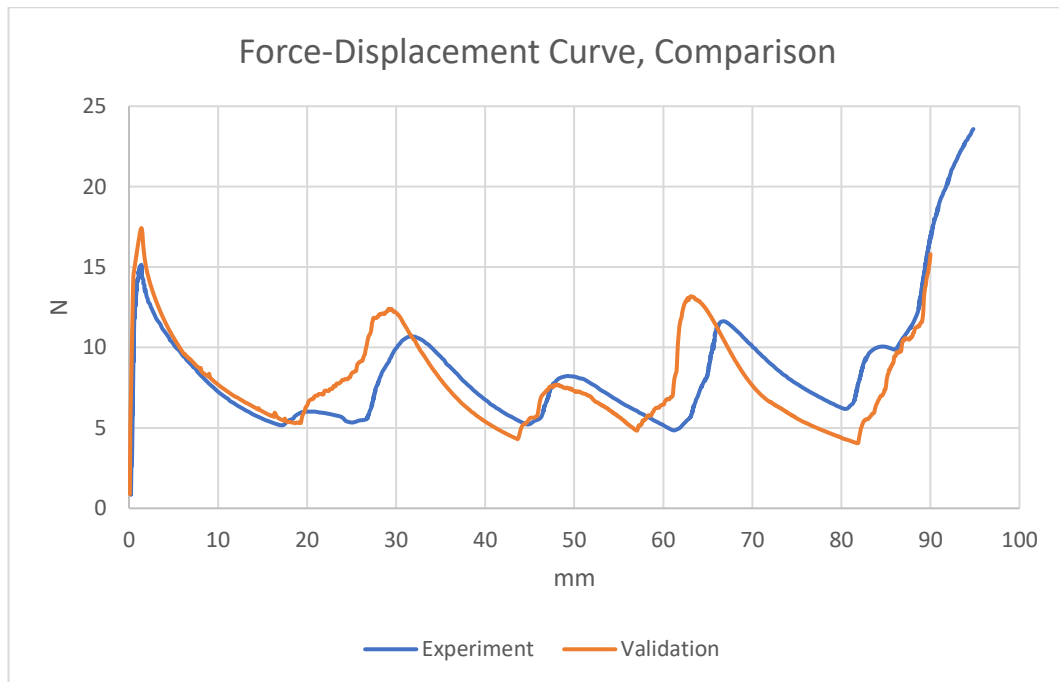


Figure 4.6 Force-displacement curve, comparison.

Fig 4.6 further illustrates the similarity between the experiment and the validation model. A total of three peaks is observed on both curves although with slightly different positions and area under the curves are also identical which indicates a comparable crashworthiness performance. The crashworthiness data for both experiment and validation model are presented in Table 4.2.

Table 4.2 Results comparison.

	Experiment	Validation	Error
SEA	11.29	11.312	0.19 %
Peak Force	15.13	17.424	13.16 %
MCF	7.37	7.389	0.26 %

Error values for the validation numerical model are **0.19 %** for specific energy absorption, **13.16 %** for peak force, and **0.26 %** for mean crushing force. The error for peak force is relatively higher because the force-displacement curves are similar in behavior, but different in value. It is concluded that these errors are present due to the lack of precise material data such as material density, Cowper-Symonds constants, and friction. Material imperfections on the specimen used in the experiment may also contribute to these errors. Based on this conclusion, the validation model is accepted and the numerical simulation parameters are verified.

4.2 Conventional Crash Box Geometries

Based on the impact simulation numerical model defined in the previous chapter and validated in subchapter 4.1, axial impact of 13.4 m/s with 8.79844 kJ of kinetic energy is subjected to three conventional crash box geometries namely a plain square tube, a 4-cell multi-cell tube, and a transverse functionally graded thickness square tube with taper ratio of 2. These conventional geometries are baseline models that is used as a benchmark for the crashworthiness performance of the transverse functionally graded thickness multi-cell crash box because these geometries have been extensively researched and applied in real world scenarios.

4.2.1 Plain Square Tube

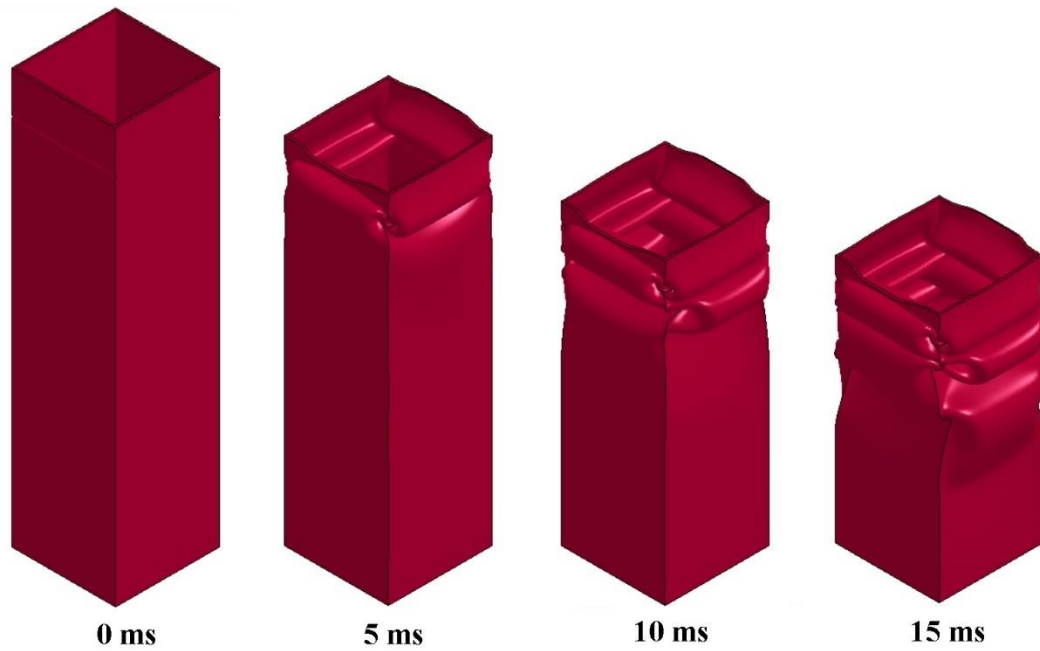


Figure 4.7 Plain square tube deformation.

The deformation mode of the plain square tube is inextensional and it started at the top region of the crash box, Fig. 4.7. The energy plot of plain square tube shown in Fig. 4.8 shows that the hourglass energy is constantly zero which means that there is no incorrectly dissipated energy. The graph also indicates that some of the kinetic energy is transformed into sliding energy, proving that the contact configuration defined during numerical modelling worked properly and contact is established.

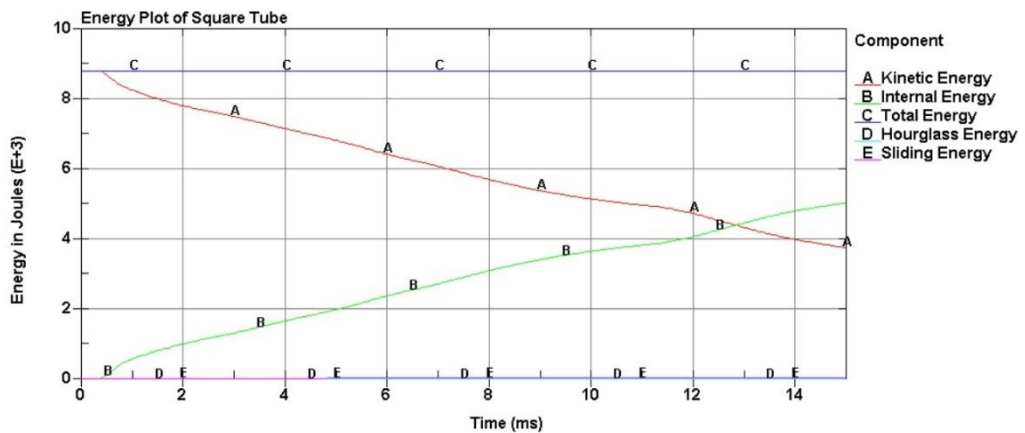


Figure 4.8 Energy plot of plain square tube.

Fig. 4.9 illustrates the force-displacement curve of a plain square tube. From this force-displacement graph, crashworthiness indicators namely energy absorbed, specific energy absorption, mean crushing force, and crushing force efficiency could be calculated and the crashworthiness performance of plain square tube could be determined.

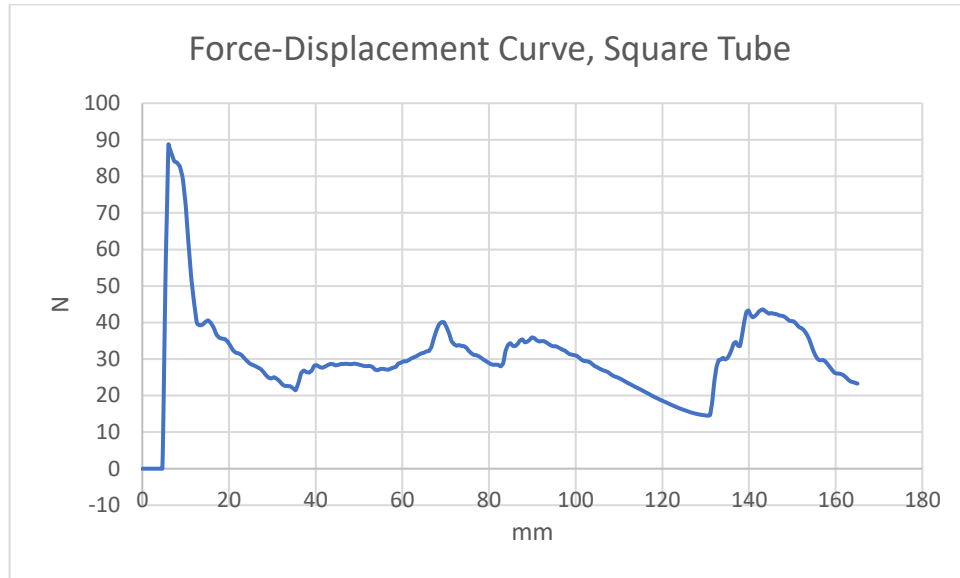


Figure 4.9 Force-displacement curve, plain square tube.

The crashworthiness indicators of plain square tube are shown in Table 4.3. These values will serve as a baseline for the transverse functionally graded thickness multi-cell crash box in order to determine whether or not the newly introduced geometrical configuration is an improvement.

Table 4.3 Crashworthiness indicators of plain square tube.

Total Specimen Mass (kg)	0.846
Max Displacement (mm)	160.04
Peak Force (N)	88.724
Displaced Mass (kg)	0.339
Energy Absorbed (kJ)	5.062
Specific Energy Absorption (kJ/kg)	14.951
Mean Crushing Force (N)	31.628
Crushing Force Efficiency	0.356

4.2.2 4-Cell Multi-Cell Tube

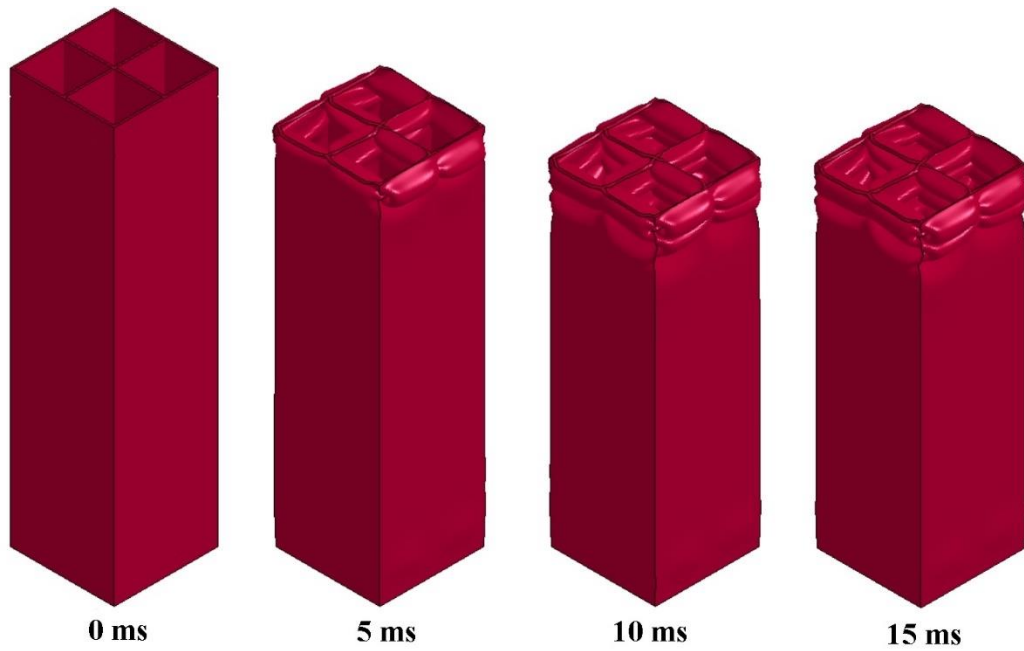


Figure 4.10 4-cell multi-cell tube deformation.

The deformation mode of 4-cell multi-cell tube is inextensional and it started at the top region of the crash box, Fig 4.10. Similar to the plain square tube, the energy plot of 4-cell multi-cell tube shown in Fig. 4.11 shows that the hourglass energy is constantly zero which means that there is no incorrectly dissipated energy. The graph also indicates that some of the kinetic energy is transformed into sliding energy, proving that the contact configuration defined during numerical modelling worked properly and contact is established.

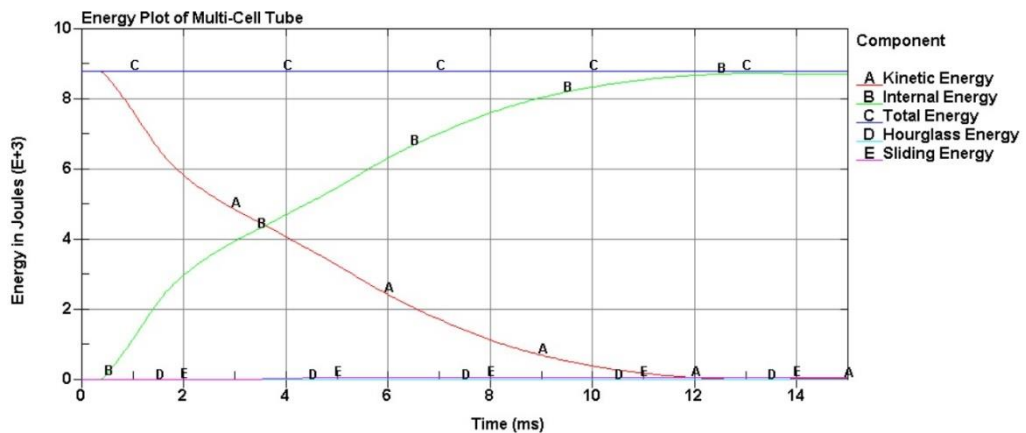


Figure 4.11 Energy plot of 4-cell multi-cell tube.

Fig. 4.12 illustrates the force-displacement curve of a 4-cell multi-cell square tube. From this force-displacement graph, crashworthiness indicators namely energy absorbed, specific energy absorption, mean crushing force, and crushing force efficiency could be calculated and the crashworthiness performance of 4-cell multi-cell tube could be determined. The crashworthiness indicators of 4-cell multi-cell tube are shown in Table 4.4.

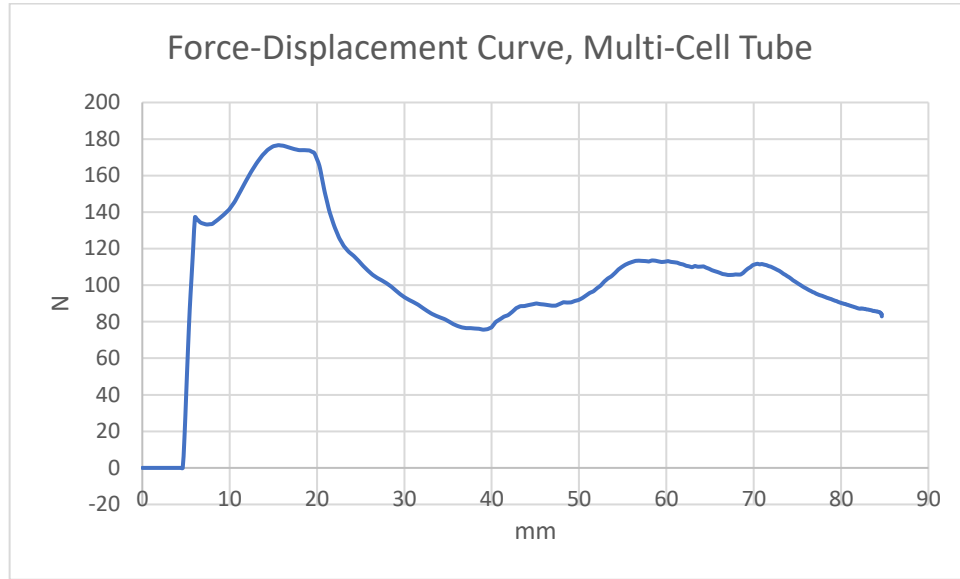


Figure 4.12 Force-displacement curve, 4-cell multi-cell tube.

Table 4.4 Crashworthiness indicators of 4-cell multi-cell tube.

Total Specimen Mass (kg)	1.262
Max Displacement (mm)	84.663
Peak Force (N)	176.639
Displaced Mass (kg)	0.293
Energy Absorbed (kJ)	8.763
Specific Energy Absorption (kJ/kg)	29.946
Mean Crushing Force (N)	94.094
Crushing Force Efficiency	0.533

4.2.3 Transverse Functionally Graded Thickness Square Tube

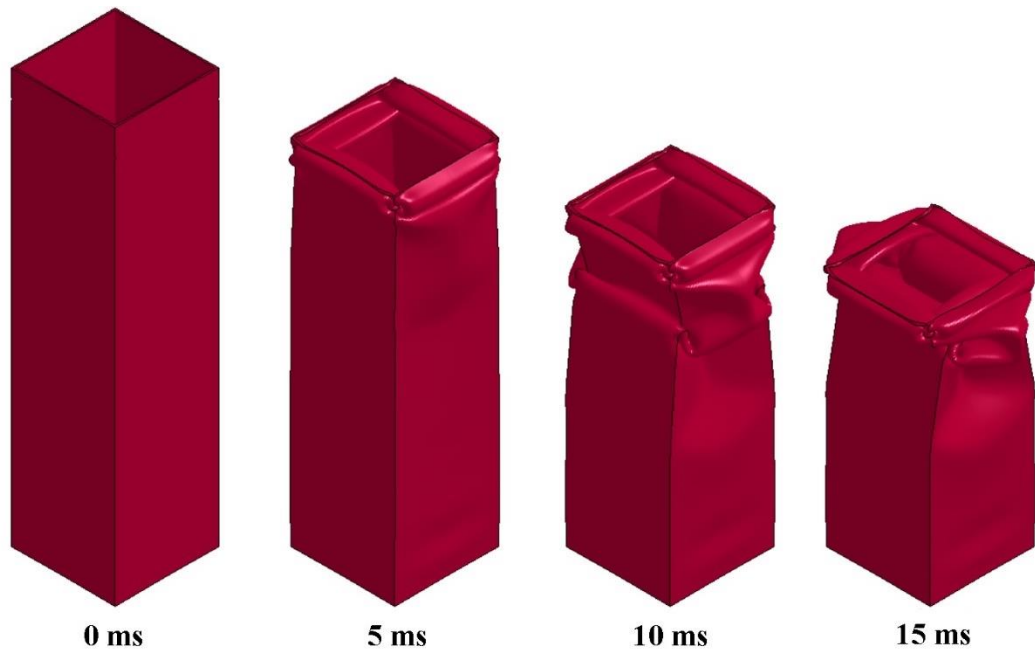


Figure 4.13 Transverse functionally graded thickness tube deformation.

The deformation mode of transverse functionally graded thickness tube is inextensional and it started at the top region of the crash box, Fig 4.13. However, it is visible that the deformation in the middle is not ideal because it does not properly form a fold. This is most likely due to the TFGT tube being relatively thinner compared to the validation model which is why the thinnest section of the tube (on the halfway point of the sides) is weak and folds vertically. However, for the sake of uniformity this result is accepted since the geometry couldn't be altered. Similar to the 4-cell multi-cell tube, the energy plot of transverse functionally graded thickness tube shown in Fig. 4.14 shows that the hourglass energy is constantly zero which means that there is no incorrectly dissipated energy and the element formulation -1 worked properly. The graph also indicates that some of the kinetic energy is transformed into sliding energy, proving that the contact configuration defined during numerical modelling worked properly and contact is established.

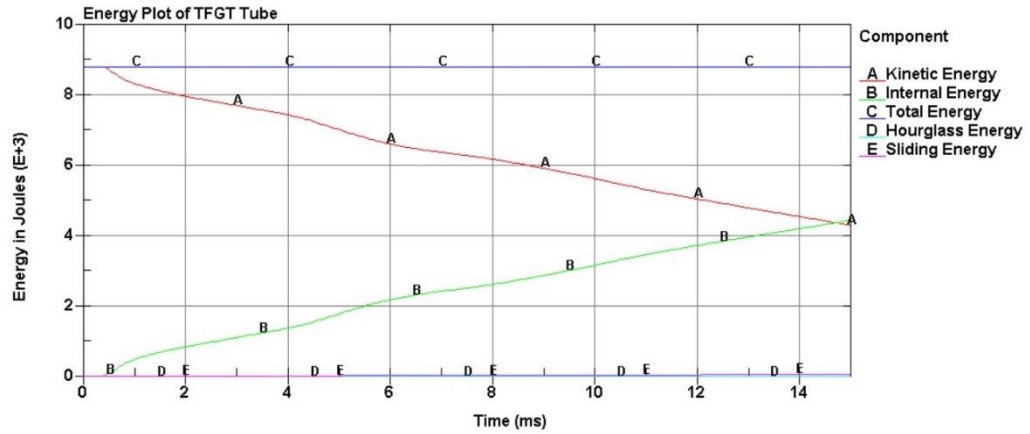


Figure 4.14 Energy plot of TFGT square tube.

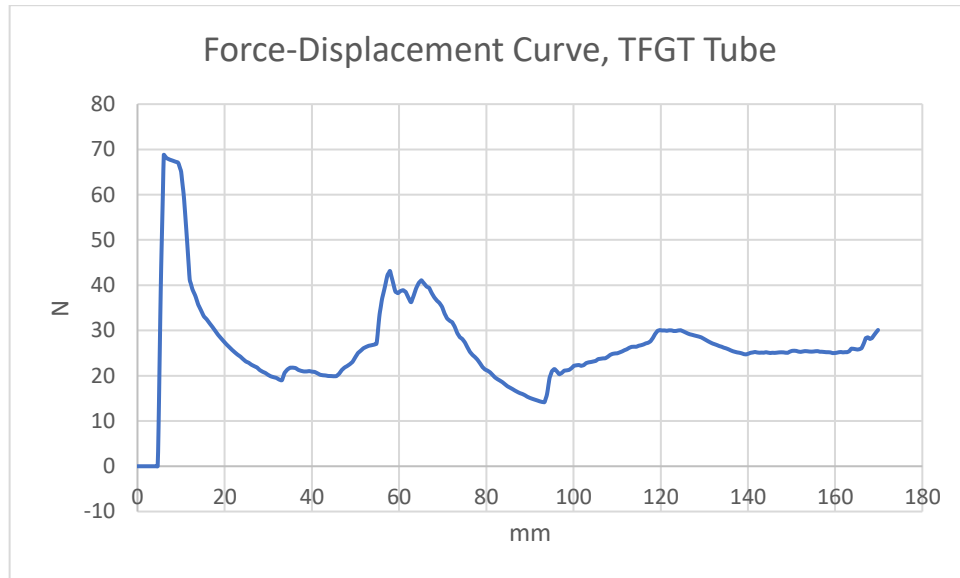


Figure 4.15 Force-displacement curve, TFGT square tube.

Fig. 4.15 shows the force-displacement graph of a transverse functionally graded tube. From this force-displacement curve, crashworthiness indicators namely energy absorbed, specific energy absorption, mean crushing force, and crushing force efficiency could be calculated and the crashworthiness performance of transverse functionally graded tube could be determined. The crashworthiness indicators of transverse functionally graded tube with a taper ratio of 2 are shown in Table 4.5.

Table 4.5 Crashworthiness indicators of TFGT square tube.

Total Specimen Mass (kg)	0.635
Max Displacement (mm)	169.84
Peak Force (N)	68.827
Displaced Mass (kg)	0.262
Energy Absorbed (kJ)	4.509
Specific Energy Absorption (kJ/kg)	17.232
Mean Crushing Force (N)	27.352
Crushing Force Efficiency	0.397

4.2.4 Conventional Geometries Comparison

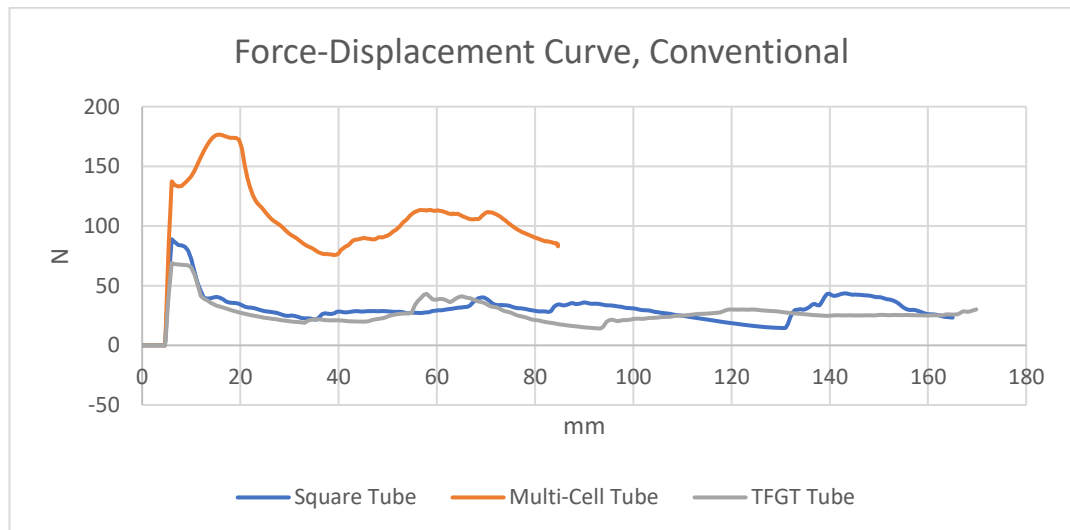


Figure 4.16 Force-displacement curve, conventional tubes.

Fig. 4.16 illustrates the force-displacement curve of three conventional tubes. It could be referred from the graph that the 4-cell multi-cell tube is the strongest and stiffest geometrical configuration because it has the smallest maximum displacement and the highest peak force. A plain square tube is the second strongest but only by a slight margin since transverse functionally graded thickness tube behaves quite similarly in its mean crushing force. The main difference between plain square tube and TFGT tube is that a plain square tube has a relatively higher peak force.

Table 4.6 Crashworthiness indicators, conventional tubes.

	SEA (kJ/kg)	MCF (N)	CFE
Square Tube	14.951	31.628	0.356
4-Cell Multi-Cell Tube	29.946	94.094	0.533
TFGT Square Tube	17.232	27.352	0.397

The crashworthiness performance indicators shown in Table 4.6 indicates that the strongest and best performing geometry between the three conventional geometries is the 4-cell multi-cell square tube. Furthermore, it is obtained from the table that although the TFGT tube is weaker as indicated by the mean crushing force, it is considerably lighter which made it the better performing geometry in terms of specific energy consumption. In addition, the TFGT tube also offers better crushing force efficiency which suggests a more stable load and a softer impact.

4.3 Transverse Functionally Graded Thickness Multi-Cell Crash Box

Axial impact of 13.4 m/s with 8.79844 kJ of kinetic energy is subjected to transverse functionally graded thickness multi-cell crash box with a taper ratio of 2 which is the baseline taper ratio for this geometry. Taper ratio of 2 is chosen as the baseline geometry because the validation model and conventional TFGT tube also utilizes a taper ratio of 2. This baseline taper ratio is used to analyze the crashworthiness performance improvement of the TFGT multi-cell crash box geometry in comparison with the widely used conventional geometries. The following deformation is obtained for transverse functionally graded thickness multi-cell crash box with a taper ratio of 2, Fig.

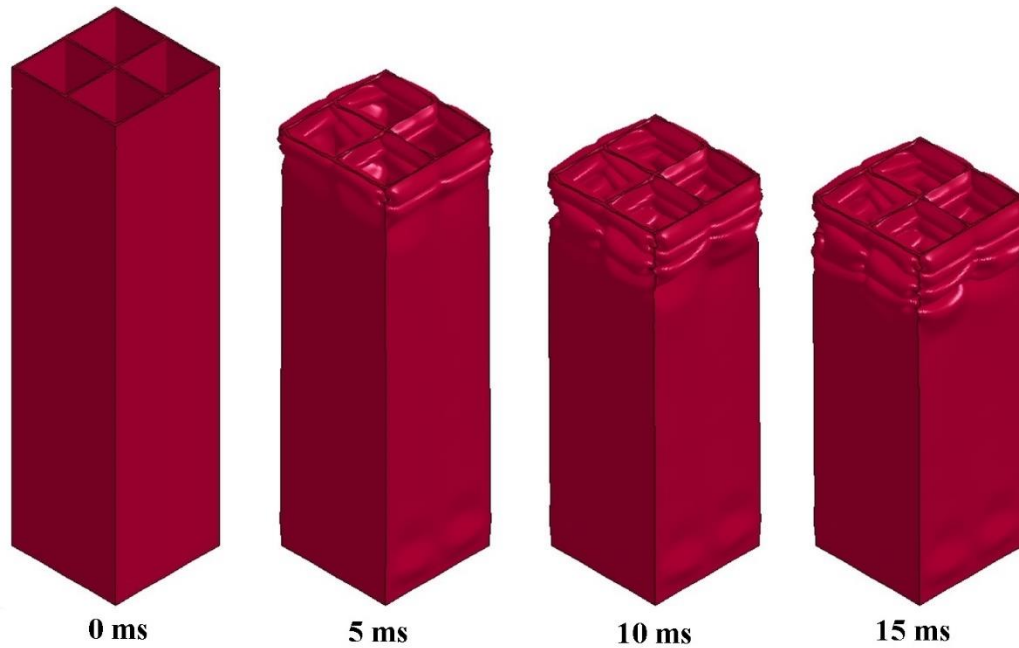


Figure 4.17 Transverse functionally graded thickness multi-cell crash box deformation.

The deformation mode of TFGT multi-cell crash box is inextensional and it started at the top region of the crash box. The energy plot TFGT multi-cell crash box shown in Fig. 4.18 shows that the hourglass energy is constantly zero which means that there is no incorrectly dissipated energy. The graph also indicates that some of the kinetic energy is transformed into sliding energy, proving that the contact configuration defined during numerical modelling worked properly and contact is established.

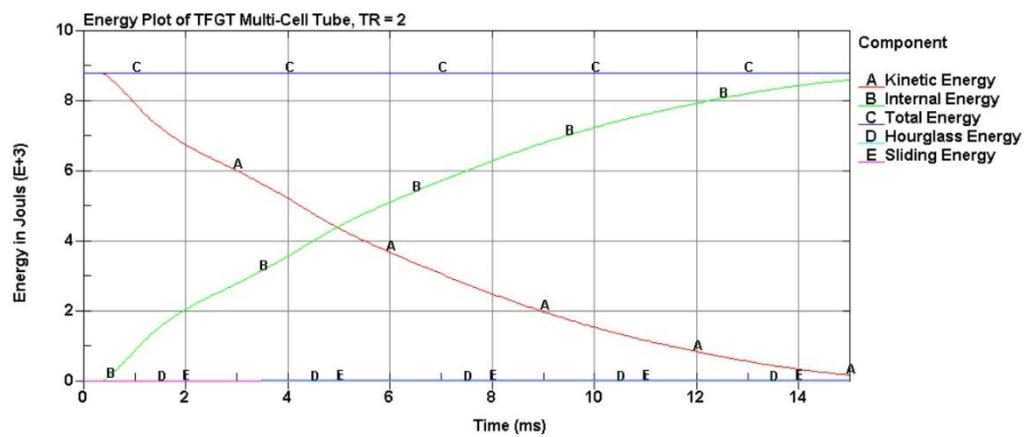


Figure 4.18 Energy plot of TFGT multi-cell crash box.

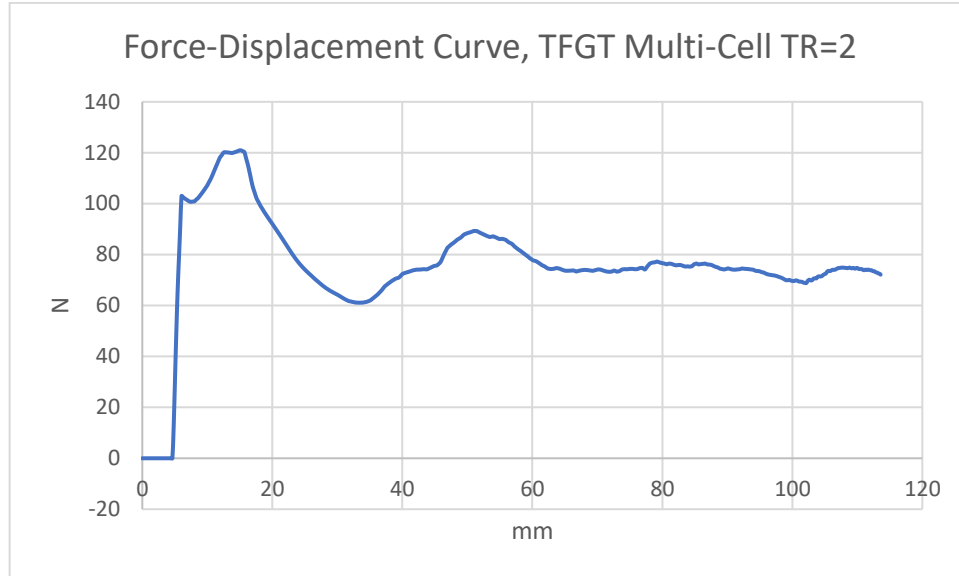


Figure 4.19 Force-displacement curve, TFGT multi-cell crash box.

Fig. 4.19 shows the force-displacement graph of transverse functionally graded thickness multi-cell crash box with a taper ratio of 2. From this force-displacement curve, crashworthiness indicators namely energy absorbed, specific energy absorption, mean crushing force, and crushing force efficiency could be calculated and the crashworthiness performance of transverse functionally graded multi-cell crash box could be determined. The crashworthiness indicators of transverse functionally graded multi-cell crash box with a taper ratio of 2 are shown in Table 4.7.

Table 4.7 Crashworthiness indicators of TFGT multi-cell crash box.

Total Specimen Mass (kg)	0.946
Max Displacement (mm)	108.58
Peak Force (N)	121.037
Displaced Mass (kg)	0.257
Energy Absorbed (kJ)	8.599
Specific Energy Absorption (kJ/kg)	33.487
Mean Crushing Force (N)	79.192
Crushing Force Efficiency	0.654

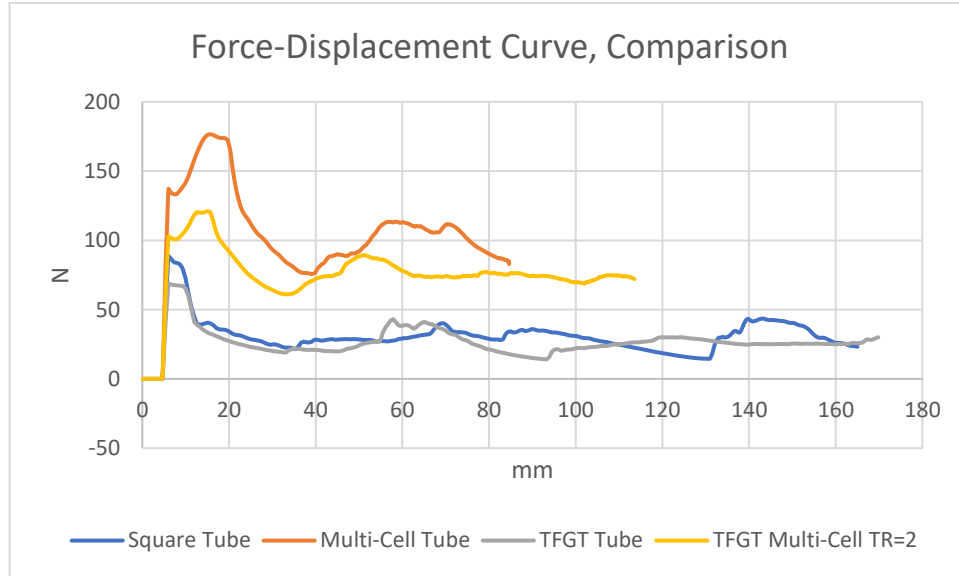


Figure 4.20 Force-displacement curve, TFGT multi-cell vs conventional.

Fig. 4.20 depicts the force-displacement curves of three conventional tubes alongside the force-displacement curve of transverse functionally graded thickness multi-cell crash box. It is visible in the graph that the peak force and mean crushing force of conventional multi-cell tube is higher. However, considering that the TFGT multi-cell crash box has a significantly lower mass, it exhibits a superior crashworthiness performance as shown in Table 4.8, 4.9, and 4.10.

Table 4.8 SEA improvement.

	SEA	TFGT-MC SEA	Improvement
Square Tube	14.951	33.487	123.978 %
Multi-Cell Tube	29.946	33.487	11.825 %
TFGT Tube	17.232	33.487	94.331 %

When compared to conventional geometries, transverse functionally graded thickness multi-cell crash box with a taper ratio of 2 has a superior specific energy absorption value with an improvement of **123.978 %** compared to conventional square tube, **11.825 %** compared to conventional multi-cell tube, and **94.331 %** compared to conventional transverse functionally graded thickness tube with the same taper ratio, Table 4.8.

Table 4.9 MCF improvement.

	MCF	TFGT-MC MCF	Improvement
Square Tube	31.628	79.192	150.386 %
Multi-Cell Tube	94.094	79.192	-15.837 %
TFGT Tube	27.352	79.192	189.529 %

When compared to the same geometries, transverse functionally graded thickness multi-cell crash box with a taper ratio of 2 has a superior mean crushing force with an improvement of **150.386 %** compared to conventional square tube and **189.529 %** compared to conventional transverse functionally graded thickness tube with the same taper ratio, Table 4.9. In contrast, the mean crushing force of TFGT multi-cell crash box is **15.837 %** lower than a conventional multi-cell tube which is most likely because it is the stiffer and stronger geometry. However, since TFGT multi-cell crash box is marginally lighter, it offers an improved performance for any other crashworthiness indicators.

Table 4.10 CFE improvement.

	CFE	TFGT-MC CFE	Improvement
Square Tube	0.356	0.654	83.708 %
Multi-Cell Tube	0.533	0.654	22.702 %
TFGT Tube	0.397	0.654	64.736 %

When compared to conventional geometries, transverse functionally graded thickness multi-cell crash box with a taper ratio of 2 has a superior crushing force efficiency value with an improvement of **83.708 %** compared to conventional square tube, **22.702 %** compared to conventional multi-cell tube, and **64.736 %** compared to conventional transverse functionally graded thickness tube with the same taper ratio, Table 4.10.

4.4 Parametric Study on Taper Ratio Variation

Further numerical simulations are conducted on transverse functionally graded thickness multi-cell crash boxes with taper ratios of 1.5, 1.727, 2.333, and 2.75 in order to determine the influence of taper ratio on the crashworthiness performance of TFGT multi-cell crash boxes. Fig 4.21 until 4.25 shows the deformation behavior of the various taper ratios with 5 ms time increment.

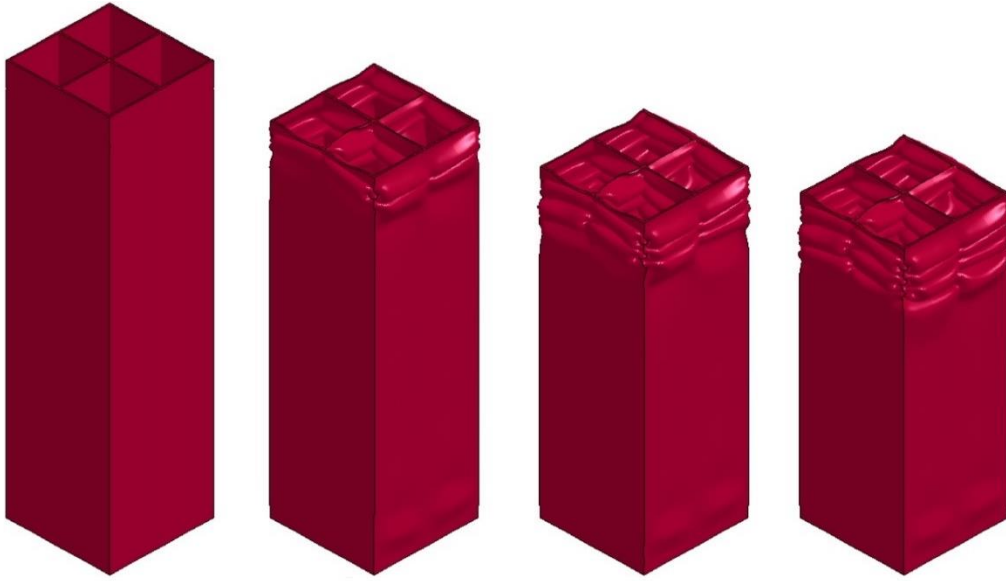


Figure 4.21 TFGT multi-cell crash box deformation, TR=1.5.



Figure 4.22 TFGT multi-cell crash box deformation, TR=1.727.

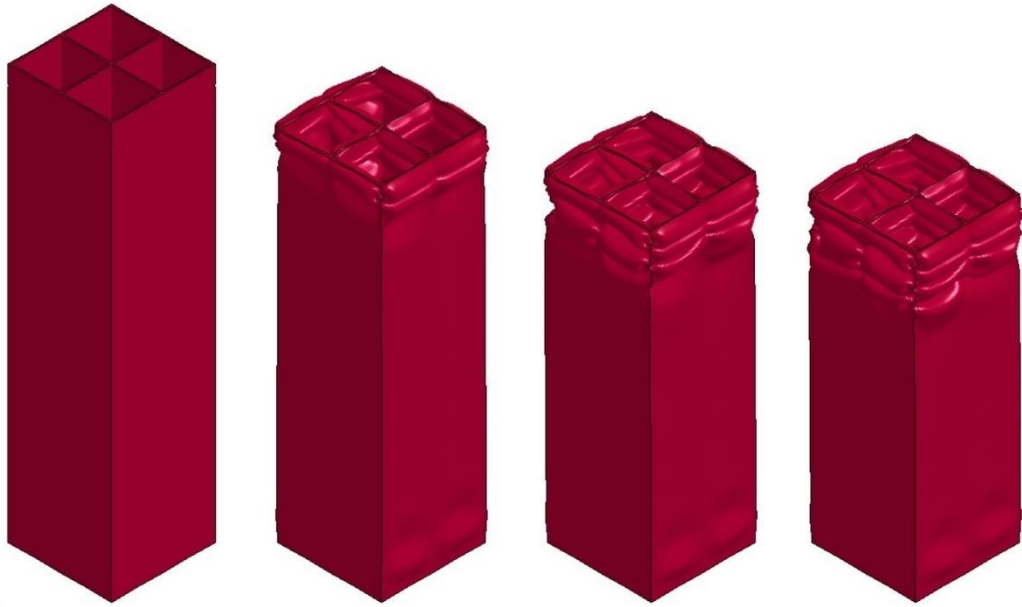


Figure 4.23 TFGT multi-cell crash box deformation, $TR=2$.

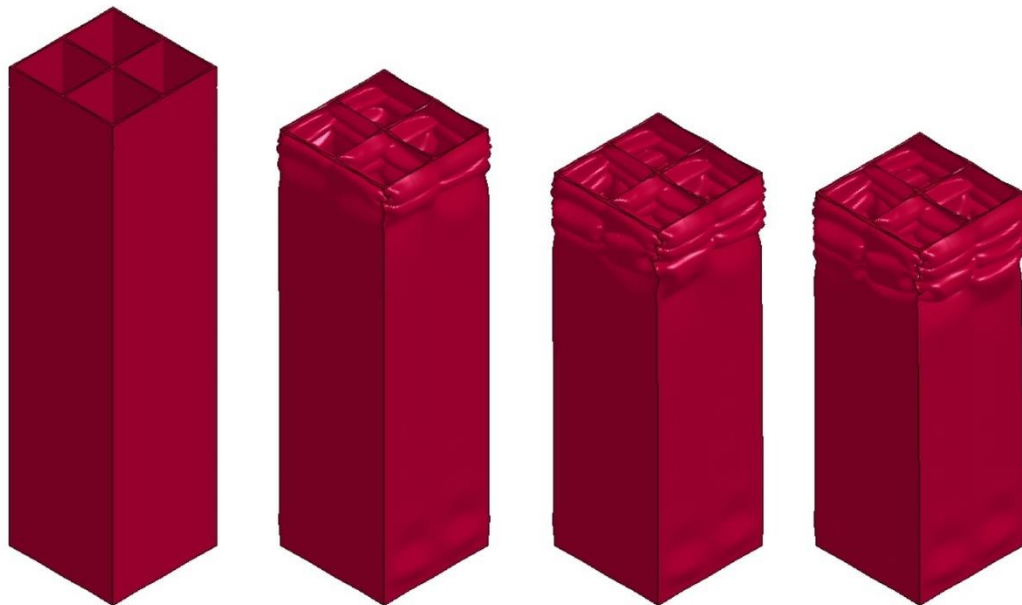


Figure 4.24 TFGT multi-cell crash box deformation, $TR=2.333$.

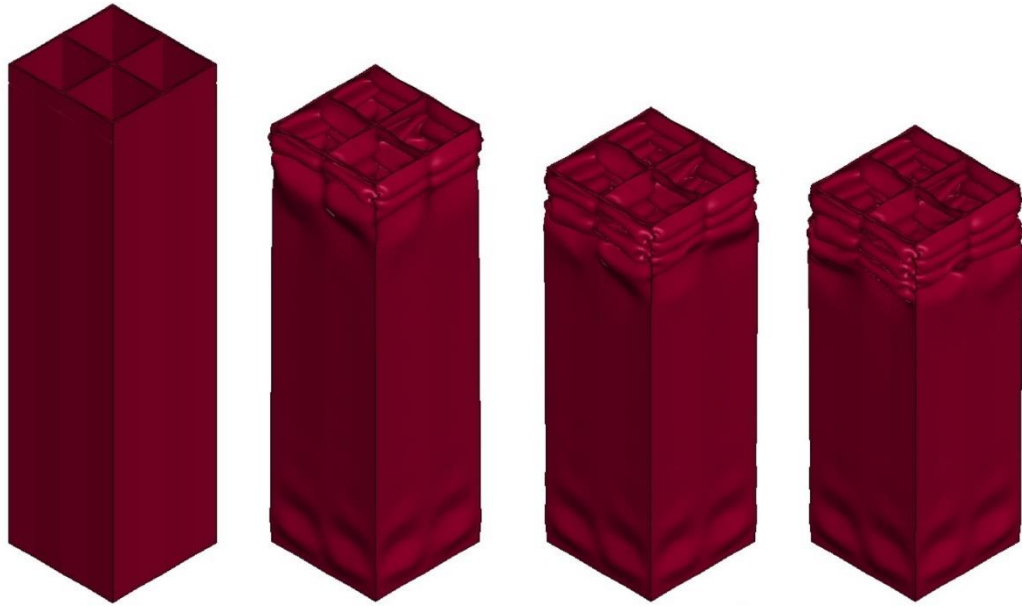


Figure 4.25 TFGT multi-cell crash box deformation, TR=2.75.

The TFGT multi-cell crash boxes similarly deform in inextensional mode starting from the top region of the crash box due to the trigger. Fig. 4.26 shows the force-displacement graph of all 5 transverse functionally graded thickness multi-cell crash boxes which indicates that the strongest taper ratio with the highest mean crushing force is 2.75 and the weakest is 1.5. The curve behavior is similar between all 5 with similar peaks and troughs.

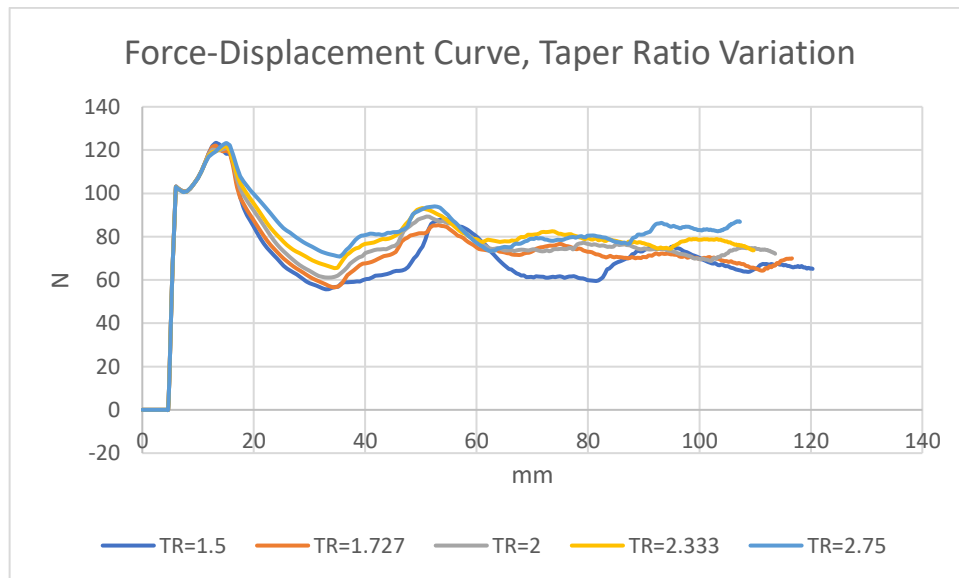


Figure 4.26 Force-displacement curve, taper ratio variation.

From this force-displacement curve, crashworthiness indicators namely energy absorbed, specific energy absorption, mean crushing force, and crushing force efficiency could be calculated and the crashworthiness performance of the transverse functionally graded multi-cell crash boxes could be determined. The crashworthiness indicators of varying taper ratios of transverse functionally graded multi-cell crash boxes are shown in Table 4.11. These values are then compared to identify the influence of taper ratio on the crashworthiness performance of transverse functionally graded thickness multi-cell crash box.

Table 4.11 Crashworthiness indicators of varying taper ratios.

	Taper Ratio				
	1.5	1.727	2	2.333	2.75
Total Specimen Mass (g)	946.225	946.015	945.935	945.985	946.163
Max Displacement (mm)	115.3	111.599	108.58	104.73	102.26
Peak Force (N)	123.428	122.179	121.037	122.001	123.336
Displaced Mass (kg)	0.273	0.264	0.257	0.248	0.242
Energy Absorbed (kJ)	8.385	8.496	8.599	8.683	8.739
SEA (kJ/kg)	30.742	32.191	33.487	35.059	36.129
Mean Crushing Force (N)	72.722	76.132	79.192	82.912	85.459
Crushing Force Efficiency	0.589	0.623	0.654	0.679	0.693

Table 4.12 SEA improvement.

Taper Ratio	SEA (kJ/kg)	Improvement
1.5	30.742	-
1.727	32.191	4.713 %
2	33.487	4.026 %
2.333	35.059	4.694 %
2.75	36.129	3.052 %
Average Improvement		4.121 %

When comparing the varying taper ratios, it is observed that with increasing taper ratio there is a gain in specific energy absorption. A SEA improvement of 4.713 % is obtained between taper ratio of 1.5 and 1.727, 4.026 % is obtained between taper ratio of 1.727 and 2, 4.694 % is obtained between taper ratio of 2 and 2.333, and 3.052 % is obtained between taper ratio of 2.333 and 2.75, Table 4.12. An average specific energy absorption gain of **4.121 %** is obtained for 5 taper ratio variations of transverse functionally graded thickness multi-cell crash box.

Table 4.13 MCF improvement.

Taper Ratio	MCF (N)	Improvement
1.5	72.722	-
1.727	76.132	4.689 %
2	79.192	4.019 %
2.333	82.912	4.697%
2.75	85.459	3.072 %
Average Improvement		4.119 %

Varying the taper ratio also exhibits an improvement in MCF. An MCF improvement of 4.689 % is obtained between taper ratio of 1.5 and 1.727, 4.019 % is obtained between taper ratio of 1.727 and 2, 4.697 % is obtained between taper ratio of 2 and 2.333, and 3.072 % is obtained between taper ratio of 2.333 and 2.75, Table 4.13. An average mean crushing force gain of **4.119 %** is obtained for 5 taper ratio variations of transverse functionally graded thickness multi-cell crash box.

Table 4.14 CFE improvement.

Taper Ratio	CFE	Improvement
1.5	0.589	-
1.727	0.623	5.772 %
2	0.654	4.976 %
2.333	0.679	3.823 %
2.75	0.693	2.062 %
Average Improvement		4.158 %

It is further observed that with increasing taper ratio there is also a gain in crush force efficiency. A CFE improvement of 5.772 % is obtained between taper ratio of 1.5 and 1.727, 4.976 % is obtained between taper ratio of 1.727 and 2, 3.823 % is obtained between taper ratio of 2 and 2.333, and 2.062 % is obtained between taper ratio of 2.333 and 2.75, Table 4.14. An average crush force efficiency gain of **4.158 %** is obtained for 5 taper ratio variations of transverse functionally graded thickness multi-cell crash box.

4.5 Sensitivity Analysis of Taper Ratio

Sensitivity analysis are conducted on transverse functionally graded thickness multi-cell crash boxes with taper ratios of 1.5, 1.727, 2, 2.333, and 2.75 in order to determine the sensitivity of taper ratio on the crashworthiness performance of TFGT multi-cell crash boxes. Fig 4.27 until 4.29 illustrates the change in value of SEA, MCF and CFE with increasing taper ratio. In general, the curve is ascending with a slight arc downwards that reduces the gradient. This indicates that the relationship between the crashworthiness indicators and the taper ratio is not linear.

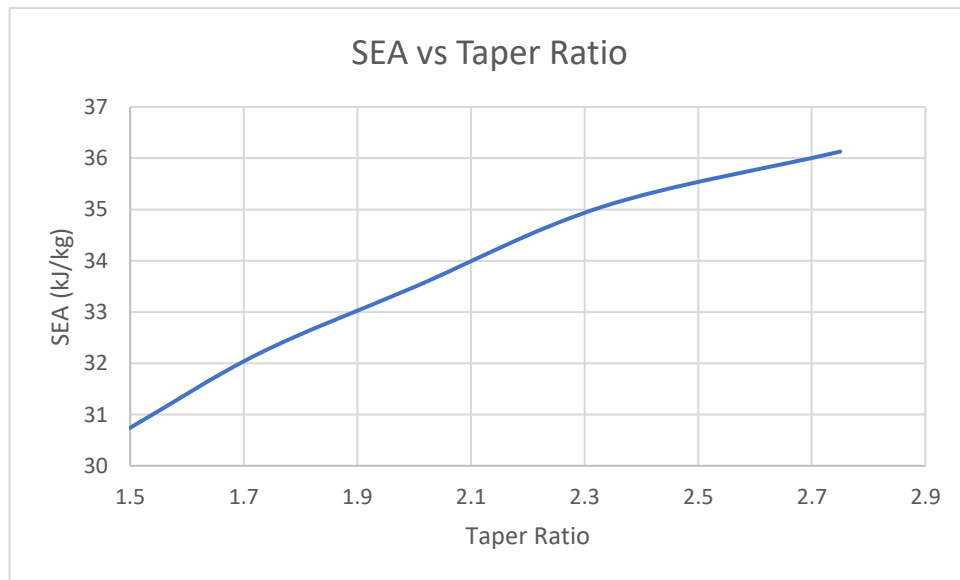


Figure 4.27 SEA vs taper ratio curve.

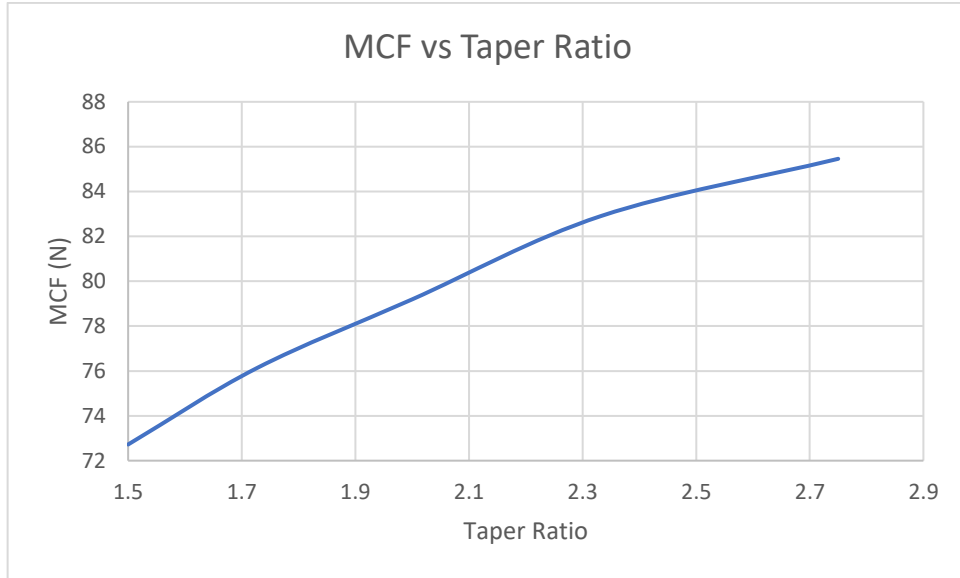


Figure 4.28 MCF vs taper ratio curve.

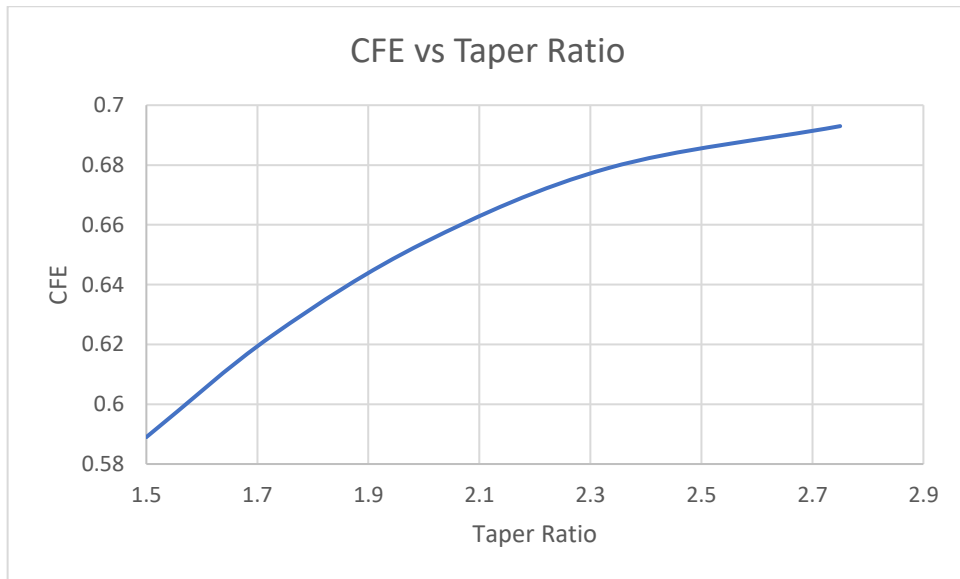


Figure 4.29 CFE vs taper ratio curve.

The value of sensitivity is calculated by dividing the change in output with the change in input. Since there are 5 variations of taper ratio (4 increments), the 4 sensitivity values will be averaged to determine the average sensitivity within the research boundaries. The sensitivity values are calculated and compiled in Table 4.15 to 4.17.

Table 4.15 Sensitivity of SEA to taper ratio.

Taper Ratio	Improvement	TR Change	Sensitivity
1.5 to 1.727	4.713 %	15.133 %	0.311
1.727 to 2	4.026 %	15.808 %	0.255
2 to 2.333	4.694 %	16.65 %	0.282
2.333 to 2.75	3.052 %	17.874 %	0.171
SEA Average Sensitivity			0.255

From a taper ratio of 1.5 to 1.727 the sensitivity of SEA is 0.311, from a taper ratio of 1.727 to 2 the sensitivity of SEA is 0.255, from a taper ratio of 2 to 2.333 the sensitivity of SEA is 0.282, and from a taper ratio of 2.333 to 2.75 the sensitivity of SEA is 0.171. Based on these values, the SEA average sensitivity to taper ratio is **0.255**.

Table 4.16 Sensitivity of MCF to taper ratio.

Taper Ratio	Improvement	TR Change	Sensitivity
1.5 to 1.727	4.689 %	15.133 %	0.310
1.727 to 2	4.019 %	15.808 %	0.254
2 to 2.333	4.697 %	16.65 %	0.282
2.333 to 2.75	3.072 %	17.874 %	0.172
MCF Average Sensitivity			0.254

From a taper ratio of 1.5 to 1.727 the sensitivity of MCF is 0.310, from a taper ratio of 1.727 to 2 the sensitivity of MCF is 0.254, from a taper ratio of 2 to 2.333 the sensitivity of MCF is 0.282, and from a taper ratio of 2.333 to 2.75 the sensitivity of MCF is 0.172. Based on these values, the MCF average sensitivity to taper ratio is **0.254**.

Table 4.17 Sensitivity of CFE to taper ratio.

Taper Ratio	Improvement	TR Change	Sensitivity
1.5 to 1.727	5.772 %	15.133 %	0.381
1.727 to 2	4.976 %	15.808 %	0.315
2 to 2.333	3.823 %	16.65 %	0.230
2.333 to 2.75	2.062 %	17.874 %	0.115
CFE Average Sensitivity			0.26

From a taper ratio of 1.5 to 1.727 the sensitivity of CFE is 0.381, from a taper ratio of 1.727 to 2 the sensitivity of CFE is 0.315, from a taper ratio of 2 to 2.333 the sensitivity of CFE is 0.230, and from a taper ratio of 2.333 to 2.75 the sensitivity of CFE is 0.115. Based on these values, the CFE average sensitivity to taper ratio is **0.26**.

CHAPTER V

CONCLUSION AND FUTURE WORKS

5.1 Conclusion

This undergraduate thesis research conducts 8 simulations of various geometrical configurations of a crash box, 5 of which is varying taper ratios of transverse functionally graded thickness multi-cell crash box. Comparison between TFGT multi-cell crash box and conventional geometries is done to establish the improvements in crashworthiness performance, while parametric study and sensitivity analysis is done to analyze the influence of taper ratio variation. Based on the results and analysis, this research has come to the following conclusions:

1. Analysis conducted on the validation model gives an error value of **0.19 %**, **13.16 %**, and **0.26 %** for SEA, MCF and CFE respectively. Based on these values alongside with qualitative assessment of the deformation and force-displacement curve, it is concluded that the validation model satisfies the requirements and is valid.
2. The deformation behavior of transverse functionally graded thickness multi-cell crash box is inextensional which started at the top region of the crash box. The deformation figures also shows that the folding pattern is very similar to conventional TFGT tube albeit there is four of them because it is a 4-cell multi-cell geometry. Overall, the deformation is satisfactory due to the fact that it successfully deforms in a stable manner and the thicker corner section doesn't negatively affect the folding sequence.
3. The crashworthiness performance of transverse functionally graded thickness multi-cell crash box with a taper ratio of 2 is **8.599 kJ** for energy absorbed, **33.487 kJ/kg** for specific energy absorption, **79.192 N** for mean crushing force, and **0.654** for crush force efficiency.
4. When compared to conventional geometries, the TFGT multi-cell tube with a taper ratio of 2 exhibits a SEA improvement of **123.978 %**, **11.825 %**, and **94.331 %** for square tube, multi-cell tube, and TFGT tube respectively. It also exhibits MCF improvement of **150.386 %**, **-15.837 %**, and **189.529 %**

for square tube, multi-cell tube, and TFGT tube respectively. Finally, it exhibits CFE improvement of **83.708 %**, **22.702 %**, and **64.736 %** for square tube, multi-cell tube, and TFGT tube respectively. The negative improvement of mean crushing force compared to TFGT multi-cell tube indicates that the multi-cell crash box geometry is still superior in terms of overall strength and rigidity. However, since the TFGT multi-cell tube is significantly lighter it offers better specific energy consumption and crush force efficiency which means that it is more weight efficient and delivers a more stable load to the passengers.

5. When altering the taper ratio of transverse functionally graded thickness multi-cell crash box, there is an average improvement of **4.121 %** in specific energy absorption, **4.119 %** in mean crushing force, and **4.158 %** in crush force efficiency. This proves that by increasing the taper ratio of the geometry, better crashworthiness performance could be achieved.
6. Sensitivity analysis conducted on the varying taper ratio configurations of transverse functionally graded thickness multi-cell crash box gives an average sensitivity value of **0.255** for specific energy absorption, **0.254** for mean crushing force and **0.26** for crush force efficiency. These values shows that CFE is the crashworthiness indicator that is most affected by alterations in the taper ratio whereas MCF is the least affected.

5.2 Future Works

Based on the results and conclusion of this research, the following are suggestions that could improve this research and serve as a guideline for future works:

1. The simulation time for the numerical models is relatively long and is the most time-consuming aspect of this undergraduate thesis research. Improvements could be made by optimizing the simulation parameters so that it could be more efficient while retaining or increasing the output accuracy.

2. Results show that applying graded thickness on multi-cell crash box results in improvements of crashworthiness performance. Further research could be done on TFGT multi-cell geometries with more than 4 cells.
3. The taper ratio variation done on this research is used only for parametric study and sensitivity analysis to establish the relationship between taper ratio and crashworthiness performance. Optimization of the taper ratio will result in the optimized geometry for TFGT multi-cell crash box.
4. Validation model of this research is based on experiment conducted on conventional TFGT single cell tube. Experiment on TFGT multi-cell tube have yet to be done and could validate the results of this undergraduate thesis research.
5. With the advantages of this geometrical configuration concluded, TFGT multi-cell tube construction could be applied to a complete vehicle numerical model or experiment to see its performance within a larger system.

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APPENDICES

Appendix A

Declaration of Authenticity of Undergraduate Thesis

STATEMENT LETTER

I, the undersigned:

Full Name : Bayu Dimas Febriansyah
Student Number : 13618121
Study Program : Aerospace Engineering
Adress/Contact : Jl. Wadas Raya, Gg. Damai, No. 98, RT 05/15, Pancoran
Mas, Depok, Jawa Barat 16436 / +6287878765930

Hereby declare that:

1. This undergraduate thesis with the title, "Impact Analysis of Multi-Cell Crash Box Structures with Transverse Functionally Graded Thickness", is an original and genuine intellectual work by me, and is not the work of others on my behalf nor it is the result of imitation or plagiarism of the work of others. This undergraduate thesis has never been submitted for an academic degree either at Bandung Institute of Technology or at any other institutions.
2. In this undergraduate thesis, there are no excerpts of writing, works, or opinions that have been written or published by others unless clearly referenced according to generally accepted academic rules, mentioning the author's name and listed in the references.
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Bandung, January 15th 2023



[Handwritten signature]

as Febriansyah

Appendix B

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